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**STRATEGIC EXAMINATION OF THE ADVANCED ENTERPRISE RESOURCE
PLANNING INDUSTRY**

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INTRODUCTION

Enterprise Resource Planning (ERP) software has been the central nervous system of business since it incorporates functions across the supply chain to finance. On-premise systems of those days were complex and were used mainly as systems of record. The industry today is at a crucial inflection point that is being driven by the transformative nature of cloud computing and Artificial Intelligence (AI). This development is repositioning ERPs as lethargic data storage systems to active, smart frameworks of strategic decision-making.

The traditional ERP system was created so that different business processes like finance, human resources, and supply chain could be fully integrated into one system, which reduced redundant data and, consequently, improved processes (Jacob and Weston, 2007). But these systems also had numerous disadvantages. They were inflexible and rigid to configure, requiring companies to spend a lot of money and time doing so. They were costlier because the systems would require outside consultants to implement and would take years (Bradford, 2015).

Moreover, conventional ERPs operate reactively and concentrate on historical data and reports rather than providing forecasting information for decisions. In those scenarios, most organizations have begun to switch to more scalable and AI powered ERP systems that can adapt in real time and provide real time intelligence.

The worldwide cloud ERP is also estimated at USD 50 billion in 2023 and is projected to have a growth rate of 13.6% CAGR through 2030 (projected USD 123.41 billion), with the AI-enabled ERP solutions having the highest growth rate in the given segment. The keynote to this tremendous expansion has been the dire necessity of the enterprises to cease relying on the legacy system and advance to the smart systems that can operate under their own autonomy. In the current state of ERP comprehension, AI-based demand forecasting, conversational interface, natural language processing, predictive analytics (to handle proactive risk management), and a smartly automated routine transaction processing are all becoming the standard features of advanced ERP systems. These features have made ERP not only an operational efficiency tool but a strategic resource providing a way to be agile as an organization and competitive.

The research is focused on an in-depth analysis of the ERP industry by the help of the application of well-known strategic management theories to see how the top three companies in the market—SAP, Oracle, and Workday—compete and even win in the realm of AI-enabled corporate systems. The critical evaluation applies Porter's Five Forces to measure industry lure and rivalry incessantly, and the VRIO model to reveal sources of indelible and temporary competitive advantages, and the core competence theory to throw light on unique skills and their imitability, and the models of business and corporate strategy to uncover differentiation approaches, vertical integration choices, diversification patterns, and alliance strategies.

The evaluation unfolds through four blended parts that systematically apply strategic frameworks. The review of previous studies identifies the progression of technology from classical ERP to AI-assisted systems and, at the same time, also presents the analytical frameworks. The section dedicated to the industry case adds empirical context through market dynamics, segmentation patterns, and detailed company descriptions. The results and conclusion section interlinks findings across frameworks to assess competitive positioning, showing how framework-based analysis concludes that SAP's wide platform integration and installed base leverage, Oracle's vertical integration and autonomous potentials, and Workday's cloud-native architecture and user experience create unique competitive positions with different implications

for sustainability. Therefore, the strategic frameworks function as analytical instruments to uncover how these companies compete, where the sustainable advantages spring from, and which resources are the most successful in the ever-changing advanced ERP industry.

LITERATURE REVIEW

Evolution of ERP Technology and Industry

Enterprise resource planning (ERP) technology has a history of over 60 years and is a trend that has been evolving with the history of computing, data handling and integration of processes in the business world. ERP was established in the 1960s as Material Requirement Planning (MRP) systems, created to compute the schedule of production and the needs of raw materials. These primitive systems only allowed control of the inventory but they were a crucial first step to linking manufacturing with business processes (Jacobs & Weston, 2007). In the 1970s, these systems were extended by the emergence of Manufacturing Resource Planning II (MRPII), which added into all these systems; capacity planning, shop floor control, and financial connections. This development marked a transition towards isolated inventory systems to systems able to make coordination of several facets of production, as an increasing managerial requirement to have a perspective on the value chain.

Later on in the 1980s, with the development of relational databases and client server computing, one could integrate various business functions into one software platform. The early ERP systems were pioneered by firms such as SAP and Oracle, they integrated finance, human resources and supply chain management into integrated systems. This transformation saved much of data redundancy and it also enabled organizations to standardize processes in various departments (Brandford, 2015). Though the traditional ERPs were quite costly, inflexible and very complex. Implementation was sometimes years of configuration and external consultants making it difficult on smaller enterprises.

Two turning points in the 1990s further increased the adoption of ERP. To start with, the Year 2000 (Y2K) issue forced businesses to upgrade their obsolete legacy systems to handle modern integrated ERP systems that could support data sensitive transactions (Devenport, 1998). Second, the pervasive commercialization of the internet enabled the vendors to web-enable the ERP systems that brought about remote access and inter-organizational connectivity. This led to the ERP II, a new generation of systems in the early 2000s that has gone beyond internal operations to suppliers, customers and partners and therefore integrates ERP into global supply chains and e-commerce networks (Rashid, Hossain, and Patrick, 2002).

The 2010s experienced a paradigm shift as cloud computing and Software-as-a-Service (SaaS) models of delivering services became popular. Cloud ERP was also scalable, subscription-based and accessible across geographies unlike the traditional on-premise ERP, which demanded large-scale infrastructure investment. This democratized the use of ERP, and small and medium enterprises now can have access to enterprise-level features. Mobile access increased the role of ERP through the support of real-time analytics and decision-making beyond the corporate headquarters (Pokala, 2024).

The inclusion of Artificial Intelligence (AI) and sophisticated analytics is transforming the ERP industry in the present decade. Current ERP vendors integrate machine learning (ML) in predicting demand, identifying anomalies and efficient production planning. Natural Learning

Processing (NLP) is the technology behind chatbots and digital assistants, making communication between the users and ERP systems more natural and conversational. Predictive analytics is a forward-looking tool that allows decision makers to take action before it happens instead of after it does by simulating and modeling possible risks (Mhaskey, 2024). The models take ERP a step further by continually learning based on historical data as well as live data, transforming the systems into more of a dynamic decision - support system rather than a fixed storehouse of information.

There are substantial disparities between conventional ERP and artificial intelligence-driven ERP. Previous generations were mainly reactive systems of record, and they had been developed to store and process historical transactions. On the contrary, advanced ERP is a proactive, adaptive and smart system capable of suggesting strategic behaviour, and automating operational changes. The intended benefits are the smaller number of mistakes in making forecasts, decreased waste in the distribution of resources, the increased resilience of the supply chain, and the accelerated reaction to modification in the market. But there are challenges associated with these benefits. An ERP powered by AI needs high-quality and large amounts of data that is always retrained, coupled with governance tools to avoid bias and errors in automated decision making (Garg, Gupta, and Dwivedi, 2023).

In general, the development of ERP can be characterized as the transformation of inventory management systems in the 1960s to cloud based, AI-driven systems that became a digital foundation of enterprise. Unlike the previous systems, which were efficiency and data consolidation orientated, state of the art ERP systems instill strategic agility as they help firms predict disruptions, optimize real time processes, and flow in and out of extended digital ecosystems.

Description of Current Technologies, Uses, and Strategic Implications

The modern ERP systems have become integrated platforms that combine AI, machine learning, and sophisticated analytics. This is being leveraged by using ML algorithms with properties of deriving complex patterns of massive datasets, thus allowing ERP systems to make the best predictions and data-driven decisions. ML algorithms are embedded in several ERP modules - including finance, supply chain, human resources, and production planning: in these areas, they process large amounts of transactional and operational information to extract complicated patterns and make predictive leads (Jawad and Balasz, 2024). ML helps the ERP to be dynamically adjusted to real-time insights, improving efficiency and versatility (Jawad & Balázs, 2024).

Instead, companies are opting to use AI-based ERP to ensure that ML models become more transparent and open to decision-makers (Jawad & Balasz, 2024). This interpretability makes the stakeholders have confidence and be able to take data-driven insights produced within the ERP environment. The ERP systems are currently able to analyze and react autonomously to incoming data through the integration of AI and help enterprises to instantly respond to shifts in demand, as well as supply conditions and market trends. Internet of Things (IoT) also enhances the capabilities of ERP to consume the real-time information provided by the networked devices, including sensors, machines in the production room, or logistics scanners, and feed them to the ML-powered ERP modules.

Over the years, studies show that ERP value creation has been modified. Reliable results of a five-year research investigated the capabilities of ERP, and found that analytics and

routinization show a slow decline without active support. These findings indicate the necessity of regular investments in the daily ERP use, intercultural teamwork, and data-driven decision-making to gain and maintain ERP value.

Uses and Applications

The applications of ML in ERP were divided into sets with great attention given to the areas of practice and industries (Jawad & Balázs, 2024), such as inventory management, demand forecasting, production planning, quality control, predictive maintenance, and supply chain management. ML models are better in terms of production schedules, making them consider factors such as equipment availability, other raw materials availability, and prices of production. They minimize resource wastage and increase the effectiveness of production (Jawad & Balázs, 2024).

The implications are far-reaching. Ciobanu et al. (2025) explain that among the four different forms of ERP system, there are implications on the parent-subsidary supply chains, e.g., the communication quality, the effects of synergy, etc. The paper has distinguished four ERP system strategies as shown in Figure 1 below: separate ERP systems, open ERP systems of different types, open ERP systems of the same type, and shared ERP systems. (Haug et al. 2010). Separate ERP Systems is when the parent and the subsidiary company have independent ERP systems with manual information transfer or partial integration between the two systems. It is useful when subsidiaries need the greatest independence, unique processes, or face regulatory/environmental differences that require separation. Open ERP Systems of diff. Types is when the parent and subsidiary have different types of ERP systems, and their data exchange is promoted. This is an appropriate method when the organizations may have systems that are well-established and yet incompatible, and thus they can collaborate without being compelled to be exactly alike. Open ERP Systems of the same type is when both entities have the same kind of ERP system, but maintain them separately, and there are open channels of controlled data sharing. This facilitates integration, standardization, and legal or operational separation where necessary. Shared ERP system is when a parent and subsidiary have a single ERP that contributes to sharing real-time data, process consolidation, and gaining tremendous synergy. This is most effective where the focus is on strict coordination, standardization and transparency, typically found in highly centralized or synergistic companies.

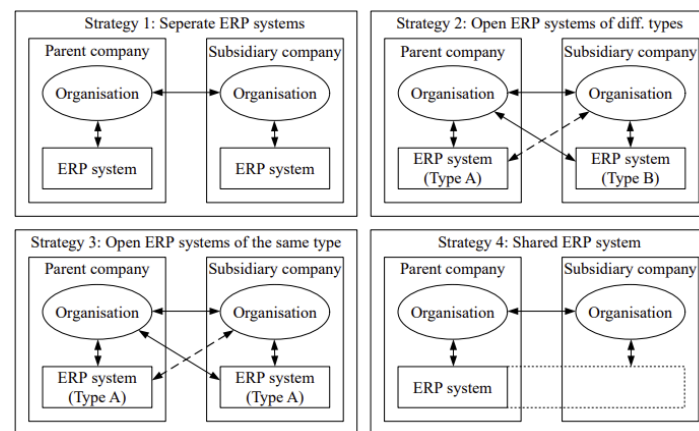


Figure 1, Four types of ERP system strategies (Haug et al. 2010)

Strategy Theories and Frameworks of Analysis

The Competitive Analysis for the Attractiveness of the Industry and the Five Forces Framework

Porter's five Forces framework is considered a foundational analysis for applying the concepts of industry attractiveness and competitive dynamics to the ERP software industry. The framework examines five forces that determine the structural conditions within an industry and its overall profit potential (Matoso & Abib, 2015). Porter's five forces are: the threats of new entrants, the intensity of rivalry between existing competitors, the threats of substitute products, the bargaining power of buyers, and the bargaining power of suppliers (Matoso & Abib, 2015). The combination of these forces provides insight into the pressures that firms have to deal with and the opportunities that can be leveraged to position the firm strategically.

This model focuses on the idea that profitability does not depend on the firm-level decisions only, but on how the industry is set up. High barriers to entry in industries, like high capital cost or high switching costs of customers, tend to keep new entrants out of the industry and maintain profitability. On the other hand, stiff rivalry in the nature of industries might impose some pressure on the margin as a result of price wars, innovation wars, and product differentiation (Jacobs and Weston, 2007; Davenport, 1998). Substitutes limit the growth of the industry by capping prices, and pricing leverage (Rashid, Hossain, and Patrick, 2002).

The Five Forces model is more of an industry-level tool, as it provides insights into external forces that apply to all firms within the same environment. However, this does not explain the differences in the performance of individual companies operating in the same industry. The latter difference is based on intra-resource, competence, and strategy. Competitive advantage is a product of the interaction between external industry forces and the firm's internal capabilities (Matoso & Abib, 2015). This two-sided approach explains why it is necessary to consider the insights of industry analysis and resource-based perspectives in the process of strategic analysis (Bradford, 2015; Ciobanu, 2025).

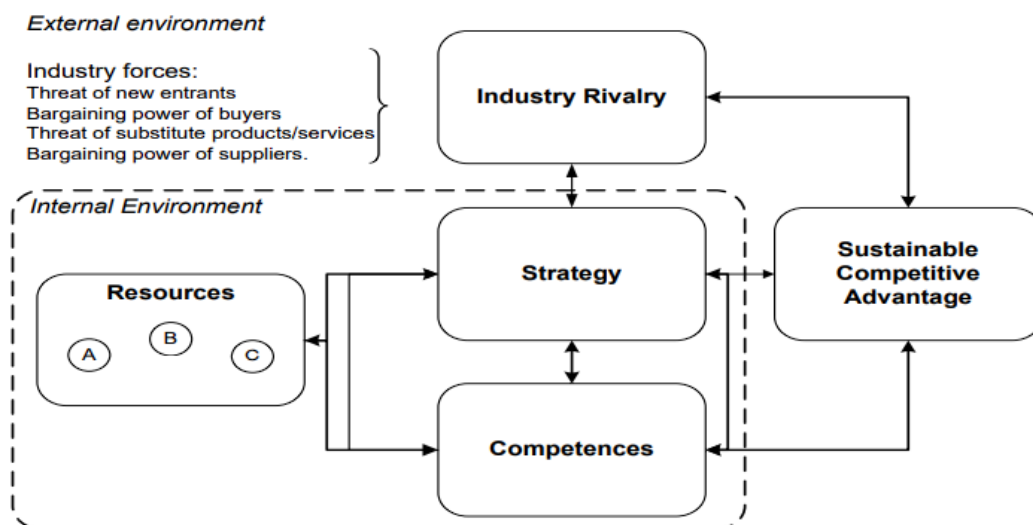


Figure 2, Relationship between categories of analysis (Matoso & Abib, 2015)

The Resource-Based View and VRIO Framework

The Resource-Based View (RBV) is a fundamental theory in strategic management that explains how firms develop competitive advantage and maintain it. The first assumption is that firms may be heterogeneous in terms of the strategic resources under their control within the same industry; the second assumption is that resources may not be perfectly mobile and transferable between firms, so the heterogeneity tends to be long-lasting (Matoso & Abib, 2015). This resource endowment heterogeneity, coupled with imperfect resource mobility, constitutes the baseline for explaining why certain firms consistently surpass others.

Consequently, there are four vital ingredients of resources that can be sources of sustained competitive advantage under the theory (Matoso & Abib, 2015). A resource can be considered valuable if it is able to exploit opportunities and neutralize threats in a firm's environment. The second differentiating characteristic should be rarity. Third, resources must be imperfectly imitable in such a way that cost disadvantages make it a significant barrier for competitors attempting to replicate competencies and resources directly. Finally, there cannot exist substitutes that are strategically equivalent to the resources and that are valuable but are neither rare nor imperfectly imitable.

The four attributes of relevance, Value, Rarity, Imitability, and Organization, are operationalized into a kind of an analytical tool called the VRIO framework for use in assessing the viability of strategic resources (Matoso & Abib, 2015). A resource or capacity, if valuable, rare, and costly to imitate, when exploited, will bring forth a sustained competitive advantage. In such situations, competitor firms find it prohibitively costly to copy the existing successful capacities and resources directly, and they cannot easily search for existing substitutes for these resources. Such resources and capacities constitute the organization's strength and represent sustainable, distinctive competencies (Matoso & Abib, 2015).

This framework further distinguishes resources that can be perfectly immobile or imperfectly mobile. Additionally, co-specialized resources can exist, which are assets that, when employed in conjunction with other resources, have high economic value (Matoso & Abib, 2015).

Most importantly, resources having all four VRIO attributes have to be exploited by the company's organizational processes (Matoso & Abib, 2015). So the organizational aspect guarantees that, through specific organizational structures, systems, and managerial practices, the potential value of resources must be drawn out. Figure 3 below explains the categories used in the content analysis by Matoso & Abib, (2015). The table categorizes the primary analytical dimensions discussed during their study that are Industry Forces, Resources and Capacities, Strategies, Organizational Competences, and Competitive Advantage, as well as the academic references that are of great importance and their subcategories. This classification demonstrates the combination of various theoretical approaches (including the industry forces and the resource-based view) to study the way companies develop and maintain competitive advantage. In this framework, the category of Resources and Capacities is the one that is directly interconnected with the above-presented RBV and VRIO framework since it is concerned with the aspects that define the performance in the long-term view.

CATEGORY	SUPPORTING REFERENCES	SUB-CATEGORIES
INDUSTRY FORCES	Porter (1986)	Competitors, Substitute products, Customers, Suppliers and New products/services.
RESOURCES AND CAPACITIES	Barney (1991)	Value, Rarity, Imitability, Organization and Substitutability.
STRATEGIES	Andrews (2006)	None.
ORGANIZATIONAL COMPETENCES	Mills <i>et al.</i> (2002)	Core, Distinctive, Organizational, Supportive and Dynamic.
COMPETITIVE ADVANTAGE	Peteraf and Barney (2003)	Financial indicators of growth and market share.

Figure 3, Categories used in content analysis (Matoso & Abib, 2015)

The core competency of the firms

Core competencies are a focal point of strategic management, which defines the overall capacity of an organization to combine and utilize resources, knowledge, and technologies to create products and services that generate a sustained competitive advantage. Those are symbolic of learning an organization as a whole, particularly in organizing and uniting the various flows of technology and expertise (Matoso & Abib, 2015).

Prahalad and Hamel (1990) argue that any capability to emerge as a core competency should satisfy three important requirements. First, it must add value to the perceptions of customers. Second, it should be distinctive and inimitable by competitors. Third, it must also facilitate the long-term expansion of the firm which will allow it to penetrate new markets or create new products in the future. These circumstances make core competencies and operational capabilities different since they are needed to be efficient but do not enable sustainable competitive advantage.

Matoso and Abib (2015) describe a hierarchy of competencies within organizations to explain how capabilities function at different strategic levels. The core competencies are the strategic capabilities that meet the requirements postulated by Prahalad and Hamel as the basis of competitive advantage in markets and products. Distinctive competencies on the other hand are competencies identified by the customers as making the firm stand out in comparison to its competitors but do not necessarily fit all the three characteristics of core competency. Organizational or business unit competencies are the competencies that are useful in certain units or operations and have a role in performance but are not always considered to be strategic resources of a firm.

The concept of dynamic capabilities, as has been addressed in the literature on strategy management, is said to be the capacity of the firm to refresh, modify and reorganize its competencies based on the environmental and technological shifts. This principle highlights that competitive advantage can only be maintained by not only having valuable competencies but also developing them as the market, technologies and customer demands evolve (Matoso and Abib, 2015).

In practice, the application of core competencies can be put into place in Enterprise Resource Planning (ERP) systems, whereby the ability of an organization to integrate processes,

data, and technology builds value over a period of time. However, this sort of value is really sustainable when the ERP-related competencies create customer value, are hard to imitate, and their continuous adaptation through dynamic capabilities emerges in line with the evolution of business needs and technological changes. Figure 4 below shows the practical application of core competencies in ERP, which demonstrates how these competencies, over time, create organizational value.

Core competencies are one of the tenets of strategic management, explaining how organizations develop and use different capabilities to develop competitive advantages. Core competencies are the capabilities of any organization to mix and integrate resources into products and services. It is essentially the collective learning in an organization, especially with respect to coordinating and integrating diverse streams of technology (Matoso & Abib, 2015). To be formally considered as competencies, three primary requirements must be met by the competencies under consideration. First, they should significantly contribute to customer perceived value. Second, they should be unique to the company and difficult for competitors to simulate. Third, they should contribute to the growth of the company in the medium and long term (Matoso & Abib, 2015). These three criteria eliminate operational capabilities from being core competencies. Core competencies are strategic assets that grant a sustainable competitive advantage.

Besides Essential Competencies, which are high-level competencies and activities at the corporate level that are essential to survival and central to the firm's strategy, the Distinctive Competencies refer to competencies and activities recognized by the customers as differentiating the firm from competitors and thus providing a competitive advantage. Organizational or Business Unit Competencies are those that are valuable to a range of competencies. Dynamic capacities are those that provide the ability for the firm to reposition its competencies in the course time, which are directly related to its resources that are most valuable to change (Matoso & Abib, 2015). This hierarchical structure acknowledges the fact that not all organizational competences adopted are to the extent that an organization performs the activities necessary for success compared to its competitors. These are the results of its capacity to combine, mix, and integrate resources into products and services (Matoso & Abib, 2015).

Core and Distinctive competencies stand as an important strategic distinction. Both bring about some sort of competitive advantage, but core competencies are said to be characterized by all three attributes described by Prahalad and Hamel (1990), while distinctive competencies differ in the market but do not necessarily fulfill all three of the core competencies attributes (Matoso & Abib, 2015). Dynamic capabilities, within this framework, deserve special attention because they are the firm's potential to renew and resuscitate its competencies in response to changes in the environmental stimulus. The contrary observation here presupposes that for companies to maintain a sustained competitive advantage, it is not merely to possess valuable competencies, but also to have the potential to evolve those competencies over time as markets, technologies, and customer requirements change (Matoso & Abib, 2015). Figure 4 below shows the practical application of core competencies in ERP, which demonstrates how these competencies, over time, create organizational value.

Type	Definition
Essential Competences	High competences and activities, at the corporate level, which are key to the firm's survival and central to its strategy.
Distinctive Competences	Competences and activities that customers recognize as differentiating the firm from competitors and which provide competitive advantages
Organizational Competences or business unit competences	Competences and key activities, expected of a firm's business units
Supportive Competences	Activity that is valuable to support a range of competences
Dynamic Capacities	A firm's capacity to adapt its competences over time. Directly related to resources that are important for change

Figure 4, classification of Organizational competencies (Matoso & Abib, 2015)

The Business Level Strategy

Competitive advantage is the capacity of a firm to realize a higher profitability in comparison with its average in the industry in the long run (Porter, 1980). The frameworks in Competitive Strategy (1980) and Competitive Advantage (1985) put forward by Michael Porter suggest that these three generic strategies, namely, cost leadership, differentiation and focus, help firms to maintain competitive advantage. These are business-level strategy, and these strategies help companies like Oracle, SAP, and Workday to position themselves efficiently in the competitive market of ERP.

The focus of the cost leadership strategy is to bring efficiency to the value chain in order to reduce the production and operational cost without compromising the quality. The company that indulges in this approach aims at achieving the economies of scale, optimization of operations and effective supplier relations at a low cost (Porter, 1985). An example is that Oracle has been seeking cost-efficiency by investing in cloud infrastructure as well as massive integration that has led to substantial operational overhead (Oracle Corporation, 2025; Goovaerts, 2025). On the same note, SAP has been automating its operations and concentrating its services on centralized platforms to ensure profitability in the face of competition pressures because of their low price offerings (SAP Investor Relations, 2024).

Differentiation strategy aims at providing distinctive value which customers view as being different and desirable. According to Porter (1985), innovation, technology, or high customer experience can be used as a way of differentiation. Differentiation used by SAP, Oracle, and Workday is based mainly on the innovation related to artificial intelligence, predictive analytics, and machine learning (Garg et al., 2023; Hayes and Downie, 2025). This has made SAP the leader in cloud ERP growth because it has differentiated itself with a robust cloud portfolio and flexibility in integrating them (Singh, 2025). The differences for the benefit of Oracle include the ability to provide data-driven automation and integrated cloud solutions, whereas the attention paid to user experience and financial management systems by Workday has enhanced its popularity (ERP Cloud Training, 2025; Reed, 2025).

The focus strategy reduces the level of competition to particular market segments or industries. Companies that use the strategy position their products based on the requirements of niche customer segments (Porter, 1980). An example is Workday, which specializes in providing

human capital and financial management solutions to specific sectors, such as education and the government, whereas Oracle and SAP have industry-specific ERP systems to manufacturing, healthcare, and retail (Apps Run the World, 2024; Fortune Business Insights, 2024). The strategic focus will increase customer loyalty and produce high switching costs based on special capabilities.

The other caution offered by Porter (1980) is that of the risk of being caught in the middle where the firm does not commit itself to any given strategy which may lead to the company losing competitive advantage. In the top tier of ERP vendors, the prevalent tendency of focusing on differentiation by way of innovation and cloud-based transformation has become the leading competitive strategy (Jawad & Balaczs, 2024; Pokala, 2024). This is the strategic position that enables companies such as SAP, Oracle and Workday to maintain long term profitability in an industry characterized by fast rate of technological changes and high customer demands.

Competitive advantage is described as the capability of a firm to exceed the average of profitability in its industry in the long run. Michael Porters framework on Competitive Strategy (1980) and Competitive Advantage (1985) suggests that firms can gain and maintain competitive advantage using three generic strategies namely: cost leadership, differentiation and focus. The strategies provide a basis of the concern of business-level strategies in every industry and especially in the advanced Enterprise Resource Planning (ERP) industry where competition is characterized by rapid technological growth and constant innovation.

The cost leadership strategy has been based on the achievement of efficiency in all activities of the value chain to reduce costs and still maintain the quality of the product or service. According to Porter (1985), this strategy helps companies to sell products at a low cost and still maintain the margins due to scale efficacy and efficiency in operations. Cost leadership may also mean building relationships with an efficient supplier, streamlining the operation of the company, and the use of technologies that lower the cost of transactions and maintenance. Nevertheless, cost leadership is important, as fixed costs of software creation, cloud infrastructure, software customization, and employee training are very high in ERP (Jawad and Balacs, 2024). Even though companies within the ERP market can implement cost-cutting strategies like cloud-based delivery solutions or software-as-a-service (SaaS) solutions, it is hard to maintain the economies of scale in the long term since innovation and product improvement requires consistent investment.

The differentiation strategy, on the other hand, is concerned in the formation of distinctiveness and provision of excellent worth to consumers. Porter (1985) points out that differentiation enables the firms to charge a high price and build customer loyalty based on innovations, design, and quality. Within the ERP industry, the aspects that differentiate technology include predictive analytics, artificial intelligence, machine learning, and adaptive automation. Such innovations turn ERP systems out of conventional administrative tools into a strategic platform that allows organisations to make real-time and data-driven decisions and take a proactive response to market changes (Jawad & Balaczs, 2024). The emphasis on differentiation through innovation enables the ERP providers to compete based on product performance as well as customer experience, integration flexibility and usability.

Focus strategy also reduces the competitive arena of the firm to the targeted market segments or industries. This may be as cost-oriented strategy, in which the company places an emphasis on efficiency in serving niche markets, or a differentiation-oriented strategy, in which the company places an emphasis on specialized solutions. In the ERP market, sector targeted or

industry-specific focus strategies are observable with regard to healthcare, manufacturing, retail, or education offerings. The ability of firms to tailor the ERP systems to the specialized operations and regulatory requirements of these industries creates a greater switching cost and better engagement with customers.

The other concept that Porter (1980) draws attention to is the pitfall of being in the middle which arises when the organization does not make decisive commitments towards one strategic direction. In the case of ERP vendors, a lack of focus in trying to maintain low costs and high innovation may be a competitive setback. Thus, there is a general inclination of the modern ERP market towards differentiation-based models, in which innovation and technological leadership become the main value-generating and sustainability generating entities (Jawad and Balaczs, 2024).

The Corporate-Level and Cooperative Strategies

Corporate-level strategy deals with general direction, scope and coordination of operations of a firm to the attainment of its long term goals. It is concerned with making decisions on which markets to venture into, the way to farm or distribute resources across business units and how to constitute value in terms of integration and synergy (Barney, 2011a). Oracle, SAP, and Workday are some of the market leaders in the enterprise software industry who have different corporate strategies to enhance their market presence and be in the growth and profitability despite a fast change in technology.

Vertical integration is a major element in corporate-level strategy and it dictates how much a company is involved in controlling its value chain activities. Barney (2011b) says that firms can also engage in backward integration, including internalizing up-stream processes like software development or cloud infrastructure, and forward integration, including extending into down-stream processes like implementation and managed services. Through the acquisition of cloud infrastructure capacity and the delivery of end-to-end software delivery platforms, Oracle has been leading the example of vertical integration and making it a full-stack provider (Goovaerts, 2025; Reuters, 2025). Instead, SAP strikes equilibrium between integration and strategic outsourcing and ecosystem collaboration that enables the company to scale and remain innovative in different industries (SAP Investor Relations, 2024). Workday follows the partial integration approach, but it keeps its own proprietary cloud technologies and works with service partners to implement and customize them (ERP Cloud Training, 2025; Reed, 2025).

Diversification is also another critical corporate-level decision, which is expansion in related or unrelated markets. Synergy is boosted by related diversification by utilizing complementary technologies and infrastructure (Barney, 2011). SAP has also expanded its products into human capital management, analytics, and supply chain solutions which are based on its ERP foundation to provide integrated business platforms (Singh, 2025). Oracle has also diversified to include database management, cloud infrastructure, and artificial intelligence applications to expand its enterprise suite (Pang and Markovski, 2024; Oracle Corporation, 2025). The expansion strategy of Workday is focused on diversification, with the company now entering the financial management and financial planning solutions based on its unified data model (Reed, 2025). Such a strategic breadth allows every company to increase customer interaction and decrease its reliance on a specific product line.

These corporate level practices are complemented by cooperative strategies that help form alliances, partnerships, and joint ventures that allow the sharing of resources and innovation

(Matoso and Abib, 2015). The result of such collaboration is that companies are able to develop value propositions without having to pay that full cost of developing them in-house. The examples of Oracle partnering with cloud providers and AI innovators, SAP collaborating with industry partners via its PartnerEdge program, and Workday collaborating with analytics and HR technology vendors show the examples of collaboration that are used to foster innovation and market penetration (Haug et al., 2010; Jawad and Balaczs, 2024). These partnerships are becoming more crucial as ERP vendors are converting their systems to ecosystems that link data, AI, and industry-specific apps.

Corporate level strategy deals with how companies can manipulate their overall direction, scope and portfolio of business operations in order to accomplish their long term goals. It entails deciding what markets to compete, the manner in which resources are to be apportioned, and how business units are to integrate their operations (Barney, 2011a). Corporate strategies play a crucial role in the advanced ERP sector to direct growth, diversification, and integration of technology as well as agility in a fast changing digital environment.

And vertical integration is also a fundamental element of corporate strategy that involves how much a company dominates its value chain. Firms may undertake a backward integration strategy by acquiring upstream services like software development platforms, cloud infrastructure or data management and a forward integration strategy toward more downstream services like implementation, support and managed solutions (Barney, 2011a). There are three theoretical perspectives analysis of the decision to incorporate vertically. Transaction cost economics school of thought focuses on alleviating uncertainty and opportunism by more closely controlling important processes. Resource based view pays attention to the exploitation of internal competencies and capabilities that are valuable, rare and expensive to copy. In the meantime, the real options school of thought promotes flexibility in investment decisions in the context of uncertainty in technology and market (Barney, 2011b). Particularly in the rapidly changing technology of the ERP industry, companies will implement partial integration, retaining in-house ownership of the proprietary technologies but outsourcing to an external partner to deliver services at scale.

The other important dimension of corporate-level strategy is diversification, which means the expansion to related or unrelated markets. Diversification enables ERP companies to grow their product line to closely related lines like human capital management, supply chain management, or customer relationship management with synergy enabled through the common data and platform infrastructure. On the contrary, unrelated diversification is an act of venturing into markets that have minimal technological overlap that is not prevalent in ERP owing to high levels of interdependence among systems (Barney, 2011b). The diversification objective in ERP industry is to increase the level of technological profundity as well as to provide integrated enterprise solutions that target various organizational functions.

Cooperative strategies are used to supplement the corporate-level strategies by focusing on the cooperation of firms. They are alliances, joint ventures, and partnerships and allow organizations to combine their resources, share experience, and speed up innovation (Matoso and Abib, 2015). Cooperation enables ERP vendors to increase their value propositions without paying the entire cost of an innovation internally. As an example, ERP companies often cooperate with cloud services vendors, developers of artificial intelligence, and industry-focused partners to increase features and enter new markets. These types of partnerships are becoming

more and more essential as ERP solutions become ecosystems that unite various technologies and service vendors.

Haug et al. (2010) also add that cooperative strategies may also be intra-parent-subsidary in nature, especially in international organizations. Parent companies might implement common ERP systems which facilitate standardization, synergy between subsidiaries, but provide local autonomy to respond to local demands. This is a balance that is efficient and not compromising on responsiveness. There are different degrees of cooperative strategies: in non-equity alliances partners do not share ownership rights, in equity alliances ownership rights exist between partners, and joint venture is a new legal entity created between partners (Barney, 2011c). All arrangements provide varying levels of control and risk and commitment. With the developed ERP industry, innovation, growth, and competitiveness are all fuelled by corporate-level and cooperative strategies. Acquisitions and partnerships and alliances allow companies to get access to the latest technologies, gain international coverage, and build strong digital ecosystems. These strategic partnerships make ERP a strategic facilitator of agility, flexibility and value-generation of the enterprise in the digital economy (Jawad and Balaczs, 2024; Haug et al., 2010).

INDUSTRY CASE

Market Size and Growth Trajectory

The Advanced ERP sector has significantly increase in the last five years mainly because there was a shift from the old on premise systems to the new cloud based architectures that are equipped with AI and ML features. The worldwide ERP software market was valued at around \$64.83 billion in 2024 and is expected to grow to \$123.42 billion by 2030 (Fortune Business Insights, 2024). The cloud ERP segment was a part of this bigger market and in 2023 it was worth \$34.8 billion and its growth rate will be 18% per year from 2024 to 2030 (Grand View Research, n.d.). Industry estimations point that cloud based ERP solutions will be responsible for about 60% of all ERP installations by 2025, which means that they will have outdone the traditional on premise setups for the first time (NetSuite, 2024).

The fastest growing part of Cloud ERP is AI and machine learning integrated with modern ERP systems. Businesses prefer smart systems with capabilities like automatic operations, prescriptive insights, and automated decision making over passive data repositories that need manual analysis. The demand is such that vendors are investing a lot in it. For instance, SAP has dedicated 15% of its revenue to AI-based R&D, Oracle has made heavy investments in autonomous computing technology, and Workday has incorporated machine learning into its entire cloud based platform (SAP Investor Relations, 2024; Oracle Corporation, 2025; Reed, 2025).

The future growth trajectory is likely to be very positive and this can be supported by a number of factors that are converging. The increasing need for integrated business management systems, the use of data analytical capabilities in real time, and hybrid work environments that require accessible systems are all factors that drive the adoption of cloud and AI. The AI capabilities are getting sophisticated at a faster rate than ever before as the vendors compete with one another in the fields of intelligent automation, predictive accuracy, and autonomous operations. Natural language processing is the driving force behind the conversational interfaces, predictive analytics is the one that does the demand forecasting and risk identification proactively, and the merchandising of systems (NetSuite, 2024) is the one that takes care of the

manual administration through self managing systems. Moreover, the compliance requirements of the regulatory authorities in different sectors are making the ERP modernization come faster as the organizations want the compliance monitoring to be embedded, the controls to be automated, and comprehensive audit trails which are more effective in advanced ERP systems than in the legacy platforms (Grand View Research, n.d.). Figure 5, mentions the cloud ERP Market size for the year 2020-2030. Similarly, Figure 6, shows the cloud ERP market share for different domains, by end use, 2023.

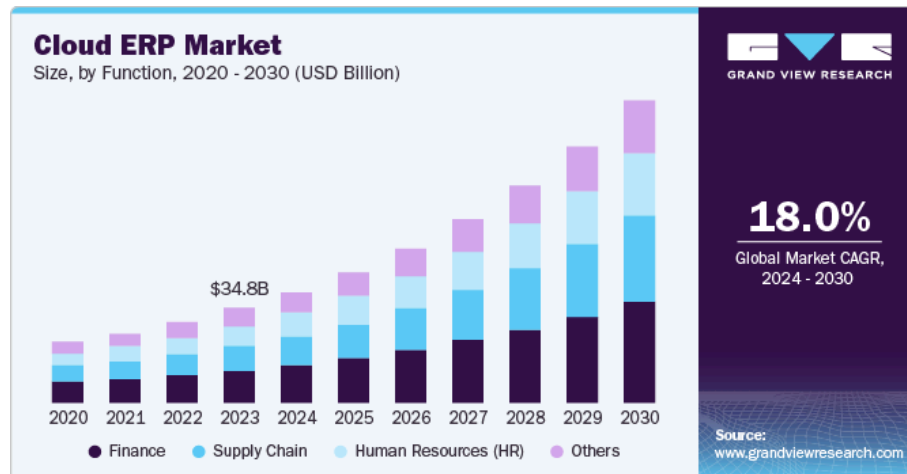


Figure 5, Cloud ERP Market Size, by Function, 2020 - 2030 (Grand View Research, n.d.)

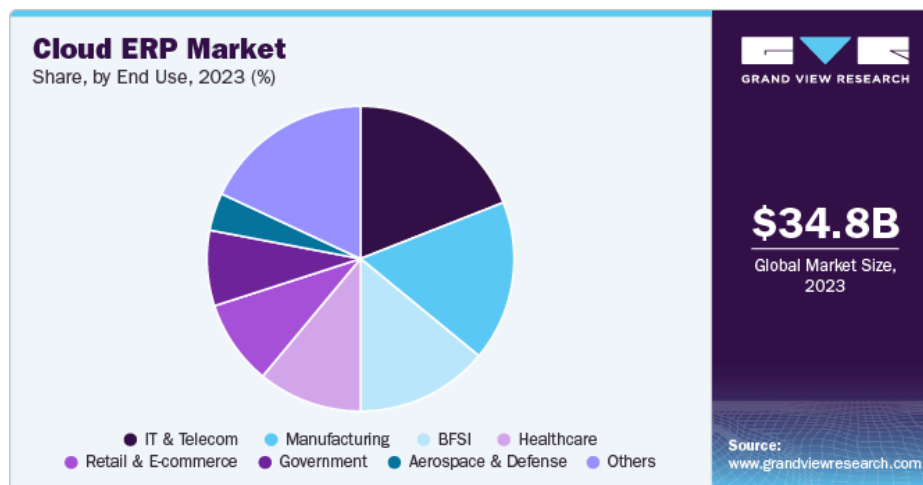


Figure 6, Cloud ERP Market Share, by End Use, 2023 (Grand View Research, n.d.)

Current Market Characteristics and Segmentation

The developed market of ERP has specific segmentation trends along various levels such as deployment models, components, enterprise size, industry segments as well as geography. Knowledge of these segments gives an idea of where growth is concentrated as well as the market categories that have the highest potential to the vendors.

Deployment Model Segmentation

Segmentation of deployment models separates the market into the concepts of public cloud, private cloud and hybrid cloud. Public cloud ERP projects dominate the market because they have cheaper first costs and shorter implementation schedules, and they are automatically scalable. Both organizations using public cloud have efficient pricing of subscriptions which saves on capital expenditure requirements and provides enterprise-grade service with shared infrastructure. The private cloud implementation is most beneficial to those organizations that have high data security concerns, regulatory limitations, or high customization demands. These deployments offer specialized infrastructure with additional control over the location of data and security settings and system changes whilst preserving the advantages of clouds such as remote access and managed infrastructure. The segment of hybrid cloud deployments is the rapidly developing category, as companies are striking the balance between flexibility and security issues. Hybrid schemes enable companies to store sensitive data and highly customized processes on-premise and utilise cloud functionality to less sensitive operations, standard processes and make them accessible worldwide (Grand View Research, n.d.).

Component Segmentation

The component segmentation is used to differentiate between the software and services in the ERP market. Software includes the licenses or subscriptions of the ERP applications such as financial management, supply chain, human capital management and industry specific modules. Services are implementation consulting, integration, customization, training, support after sales, and managed services. Services segment indicates a high market value due to the fact that most ERP implementation projects involve substantial levels of professional services in the form of requirements analysis, business redesign, data migration processes, systems configuration, integration with existing applications, and user training. The cost of services of enterprise implementations may be greater than the cost of software, particularly of complex global implementations where a large amount of customization and control of change may be needed (Grand View Research, n.d.). Figure 7 shows the ERP Customer Distribution by Vendor and Organisation Size (HG Insights, 2025).

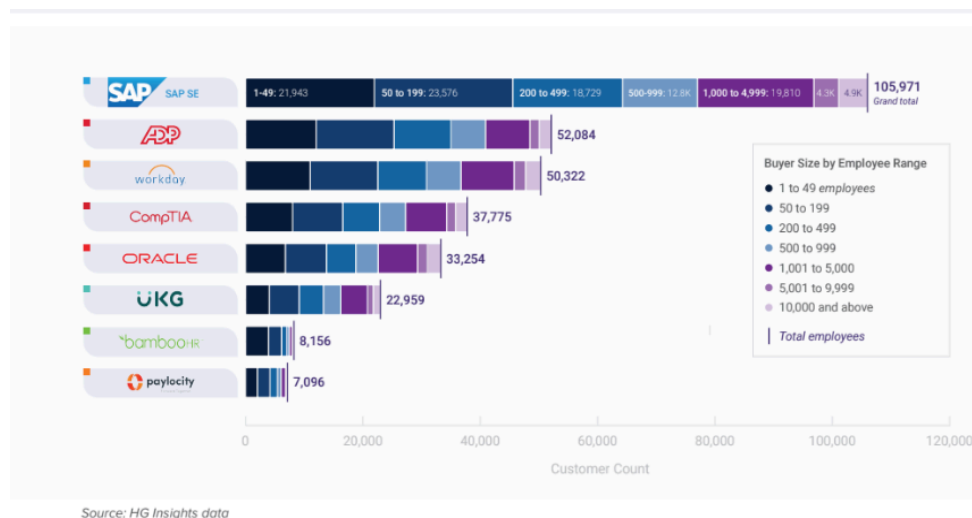


Figure 7, ERP Customer Distribution by Vendor and Organisation Size (HG Insights, 2025)

Enterprise Size Segmentation

Enterprise size segmentation is used to divide the market into small, medium, and large enterprises in terms of the number of employees, revenues, and complexity of the operations. Big companies now occupy the majority of the market share because they have complicated requirements to operate involving various business units, geographic regions, and functional areas. These companies demand elaborate ERP packages having ample functionality, worldwide implementation as well as advanced integration with many additional enterprise systems. They have large IT budgets to run multi-million dollar implementations and maintain it (HG Insights, 2025). Small and medium enterprises are the most rapidly expanding category since vendors are proposing accessible pricing frameworks, module-based structures, and business-centric templates decreasing implementation intricacies and overall cost of possession. Cloud delivery models will particularly assist SMEs as they do not involve the need of investing in the infrastructure and the price of subscription is dependent on the size of the business. This is coupled with an increase in targeting this segment with simpler implementations, industry packages and lower priced packages so that organizations with a hitherto inaccessible level of Enterprise-level ERP can now afford to access it (HG Insights, 2025).

Vertical Industry Segmentation

Vertical segmentation in the industry indicates the adoption patterns of industries in manufacturing, retail, healthcare, banking and financial services, information technology, telecommunications, and others. The needs to optimize the supply chain, plan production, control quality, manage inventory and comply with regulations continue to lead to the adoption of ERP in manufacturing. ERP manufacturing also includes specific bill of materials management, production scheduling, shop floor control, and intricate supply chain coordination functionality. The market leadership of the sector can be attributed to its early adoption of ERP and its further modernization to include the use of AI (Grand View Research, n.d.).

Healthcare reflects an accelerating ERP adoption rate based on regulatory compliance demands, operational efficiency demands, and financial transparency needs. Healthcare organizations use ERP to manage finances, optimize the supply chain, schedule workforce, and report on regulations as they integrate with specialized clinical systems. Retail is also increasing with a good growth because organizations are seeking to integrate their online and physical stores into omnichannels and also manage their inventory to optimize stock in distribution channels and also improve their customer experience through single commerce platforms. The banking and financial services use ERP to consolidate financials, regulatory reporting, risk management and efficiency in operations (Grand View Research, n.d.).

Geographic Segmentation

Geographic segmentation includes North America, Europe, Asia-Pacific, Latin America and Middle East and Africa. North America is the largest regional market based on its early adoption of technology, availability of large ERP vendors, enterprise IT expenditure, and the availability of advanced cloud infrastructure to support advanced deployments. The high-scale businesses in the region are the first to adopt AI-enabled ERP since they are interested in creating competitive advantages by smart automation and intelligence to predict (Grand View Research, n.d.).

The second-largest regional segment is Europe, which has witnessed growth due to compliance requirements of the General Data Protection regulation, which has pushed many

industries to modernize their systems, initiate digital transformation, and high manufacturing base that needs advanced ERP features. Asia-Pacific shows the greatest growth rate with the acceleration of industrialization, government digitalization programs, growing uptake among SMEs as well as diversified manufacturing activities in both China, India and the Southeast Asian nations. There is an emerging growth in Latin America and Middle East and Africa as economic growth and investment in digital infrastructure continue (Grand View Research, n.d.).

Market Share

The market share analysis is a study that focuses on the competitive statuses of particular vendors in the advanced ERP market. SAP leads the largest market share estimated at 24% which demonstrates its long term market dominance, diverse product range and high implementation in big enterprise ventures especially in the manufacturing and retail sectors (HG Insights, 2025). Oracle is estimated to have 12 percent market share, and it has experienced rampant growth in the recent past due to vigorous cloud expansion policies and competitive positioning of autonomous ERP facilities. It is expected that Oracle can threaten the total market dominance of SAP due to the rising speed of cloud adoption (Pang and Markovski, 2024; Zwets, 2025).

Workday is one of the most rapidly expanding vendors, having about 8% market share where the strength lies in the human capital management with financial ERP. The ability to be deployed on the cloud and the current user experience is especially appealing to the organizations that need to upgrade their old systems with cloud-native systems (Reed, 2025). Microsoft Dynamics is still with strong market shares especially in the middle market where they have integration opportunities with the entire Microsoft ecosystem that offers them competitive advantages. Specialty vendors, such as Infor, NetSuite, Sage, Epicor, and others, are considered to hold a large market share, with narrow industry verticals, geographic market segments, or enterprise size segments all having specific solutions (HG Insights, 2025).

The competitive environment is moderate with a concentration of the leading vendors with large shares but many other vendors with a niche segment. The market share dynamics change because cloud transition benefits go to vendors who have better cloud-native architecture or cloud migration strategies. The speedy expansion and huge subscriptions of Oracle to the clouds indicates that there may be a market share increase whereas the sizable base of installed software in SAP offers the defenses even when they venture into the clouds. The growth path of Workday suggests further expansion of market shares especially among businesses that have been moving towards the adoption of legacy financial systems. The market share allocation implies the opportunities of vendors based on the difference in AI functions, industry focus, or better user experiences (Pang and Markovski, 2024; Zwets, 2025).

Industry Main Players

The current strategic analysis considers three industry leaders based on their market positions, technological leadership through AI enabled ERP, and varied strategic numbers: SAP SE, Oracle Corporation, and Workday, Inc. These three companies together hold around 44% of the total ERP market share, and they are different in their ways of developing advanced ERP products. SAP goes for installed base migration with full AI integration, Oracle stresses infrastructure first vertical integration with self operating computing, and Workday is using the approach of cloud native disruption through the integration of machine learning (HG Insights, 2025).

ERP industry Market share

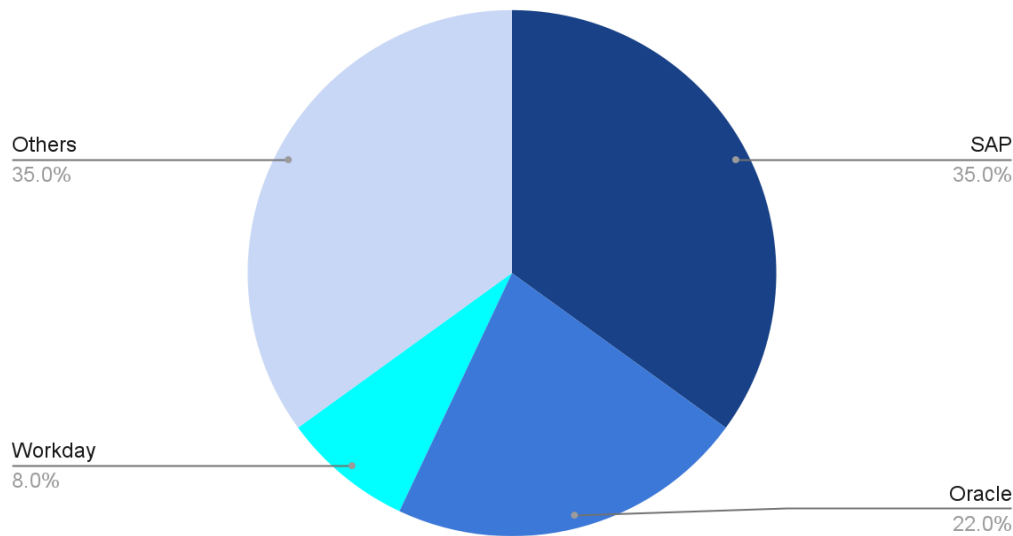


Figure 8, Global ERP Market Share by Vendor (2024).

Figure 8 represents the estimated market share of the top Enterprise Resource Planning (ERP) suppliers by 2024. SAP holds the highest position with a figure of around 35% and the next position goes to Oracle with a figure of around 22% and finally Workday with 8%. The remaining 35% is shared by vendors like Microsoft, Infor, Epicor, Sage, etc.

SAP SE - SAP has established its position as the world's leading ERP provider, having a substantial market share of 24% and recording revenues of €8.47 billion in the third quarter of 2024, which corresponds to a rise of 9% compared to last year (SAP Investor Relations, 2024). The firm has a customer base counting 440,000 in 180 countries and covering 87% of the Forbes Global 2000 companies, in addition to about 110,000 employees worldwide (SAP Investor Relations, 2024). SAP has been devoting around 15% of its annual income to R&D, which amounts to over €1 billion directed toward the integration of AI, development are the incorporation of AI in S/4HANA cloud for intelligent automation, creation of the Joule AI assistant for interactive interfaces, and enhancement of real time AI processing through in memory computing technology (SAP Investor Relations, 2024).

SAP has a huge pile of patents concerning the in memory database architectures aiming at AI workloads, enterprise AI applications for predictive analytics, and intelligent automation along with machine learning algorithms for business process optimizations. Instead of construction an infrastructure of its own, the company goes through partnerships with Microsoft Azure, AWS, and Google Cloud which gives it the liberty to concentrate on the application layer AI development. The S/4HANA cloud application has in its core AI solutions for intelligent cash application and predictive planning in the financial management area, demand sensing and predictive maintenance supply chain management, and spend analytics and supplier risk assessment in procurement (SAP Investor Relations, 2024). The HANA in memory database allows for the performance of both transactions and AI workloads to take place at the same time without any degradation of performance thus giving the real time predictive insights that are not possible in traditional database architectures.

Oracle - It is located in the second position for global ERP vendors with about 12%, its market share and \$13.3 billion total revenue in fiscal Q1 2026, the growth of which came from 10% in cloud services (Oracle Corporation, 2025). Additionally, the organization has a case of over \$500 billion in cloud orders and more than 160,000 employees worldwide (Reuters, 2025). Oracle dedicates a lot of resources to the development of autonomous computing technologies, AI integration throughout the technology stack, and the establishment of proprietary cloud infrastructure which involves investments worth billions of dollars spread across several years. The R&D in ERP of the company is advanced by autonomous capabilities that will produce self tuning, self patching, and self repair systems throughout infrastructure and applications, AI models for financial automation, and vertical integration optimizing AI performance (Oracle Corporation, 2025).

The patent portfolio of Oracle includes technology for autonomous databases, features for applications powered by AI, techniques for vertical integration, and security automation. Oracle, unlike its competitors, takes advantage of the infrastructure of its proprietary cloud which is optimized for its workloads, has fewer third party dependencies, and has architectural control allowing autonomous capabilities across all levels of technology (Oracle Corporation, 2025). Oracle fusion cloud ERP stands out with its autonomous characteristics that let its database performance auto tune itself, patching done without downtime, self repair and resource scaling done intelligently. Integration of AI brings in anomaly detection in financial transactions, automated journal entries, intelligent cash positioning, predictive reconciliation, and procurement intelligence. The vertical integration allows the best performance because the infrastructure, database, and applications are working as a whole instead of separate components (Oracle Corporation, 2025).

Workday - Workday, with 8% market share, has become the fastest-growing major ERP vendor. It has announced its fiscal 2024 revenue of \$7.26 billion, which is a 17% increase compared to the previous year's revenue (Reed, 2025). The company has more than 10,500 customers worldwide and has a good footing in Fortune 500 companies, educational institutes, healthcare, and professional service firms, where it has about 18,500 employees (Reed, 2025). Workday is making a big investment in Research and Development that is solely dedicated to cloud based application development with AI that is being used throughout the process, and the company is also prioritizing machine learning for all applications, having natural language processing for conversational interfaces, predictive analysis, adaptive planning, and continuous user experience innovation.

Workday holds various patents, which include the cloud native multi-tenant architecture, the unified HR finance data models, the application of machine learning in talent analytics and financial planning, and the usage of natural language processing in the enterprise (ERP Cloud Training, 2025). The company has an alliance with AWS for infrastructure instead of constructing its own data centers, thus operating a multi-tenant system where all the customers have the same version with the continuous deployment of AI (Reed, 2025). Workday Financial Management and Human Capital Management make heavy use of AI such as smart financial Planning, analytics of spend, predicting attrition modeling, interfering skills, matching of the talent marketplace, and overseeing compliance automatically. The Unified data model brings together HR and finance data making it easier for machine learning to detect cross-functional patterns that connect workforce costs to financial outcomes, which separated systems cannot do (Reed 2025; ERPcloud Training, 2025). Natural language processing provides the ability to have

a conversation, thus lessening the demand for training and making a difference in the user experience as compared to the conventional ERP interfaces.

General Environment Affecting the Industry

The ERP industry is impacted by technological, economic, regulatory and social causes that directly affect strategic decision, market forces and customer adoption process. The fast pace of technology change is radically transforming ERP capabilities, deployment patterns, and competitiveness. The maturity of cloud computing infrastructures among hyperscale cloud providers (AWS, Azure,Google cloud) has reached a level of infrastructure sophistication, and with deployments of ERP networks featuring high availability (99.99%+ uptime), global scalability, and infrastructure security that compares to or surpasses on-premise deployments (Panorama Consulting, n.d.). This maturity in infrastructure has removed technical obstacles to the implementation of cloud ERP and has shifted the emphasis to business processes, organizational change, and value generation.

The new technology in AI is generative AI, in which sellers are experimenting with use cases involving automated content generation, code generation and intelligence assistance assisting business processes execution.(NetSuite, 2024). The integration of edge computing and Internet of Things (IoT) expands the capabilities of ERP to operational technology spaces, allowing it to collect real time data on manufacturing devices, logistics sensors, retail point of sale systems, and asset monitoring devices. Blockchain technology is still new, however, it offers supply chain visibility, automated contracts, intercorporate exchanges, and management of digital assets (Panorama Consulting, n.d.).

The global economy has a significant impact on the different patterns of investment in ERP, its deployment, and the performance. During periods of economic growth, companies intensify their digital transformation and ERP modernization projects, perceiving technology investments as key drivers of growth and competitive advantage (Grand View Research, n.d.). On the other hand, economic downturns tend to postpone discretionary IT investments, but any project with a justified ROI will be pursued. The movement to cloud ERP and its subscription based pricing structure has made ERP more affordable during periods of economic uncertainty by reducing initial capital expenditures requirements and allowing for a more incremental implementation strategy. The level of interest rate impacts the financing expenses of a large scale ERP implementation, preferences in payment terms, lease-versus-buy, and general project authorization challenges. Currency changes affect the ROI of multinational companies when standardizing their ERP systems globally, as the system may be implemented using local currencies, as well as the projected benefits of business process standardization and global visibility (Grand View Research, n.d.).

The workforce has an expectation that their company's ERP software feels like an easy to use, mobile friendly and seamless navigation app. Remote and hybrid work environments require an ERP platform that can be accessed remotely, ensuring strong security and user experiences that can be performance based and not limited geographically. It has been found that organisational aspects, such as change preparedness, executive sponsorship and user training, have a significant impact on the success of ERP's more than technical aspects (Panorama Consulting, n.d.). This fact has driven the vendors to make investments in implementation methodologies, adoption analytics, and change management resources. This shift in the generational workforce proposes different degrees of digital literacy and expectations of technology, where young workers are likely to demand easy-to-use interfaces, mobile-first

interfaces, and direct links with collaboration tools, while older workers may need to work with what they already understand and adapt slowly.

Industry Trends

The advanced ERP industry is evolving along several strategic trajectories that define where vendors and customers are heading in terms of technology architecture, delivery models, market positioning, and ecosystem relationships.

Companies are moving to cloud from on premise systems, but the transition is a critical process. The manufacturers are no longer providing support for the old system due to this technology transition. As an example, SAP plans to stop support for ECC by 2030; meanwhile, the company is charging higher fees for on-premise maintenance, which is making the cloud option even more attractive (SAP Investor Relations, 2024). Nevertheless, the path customers are taking is not entirely in the direction of the public cloud, as hybrid variants are becoming more and more popular. In fact, companies are designing their infrastructure in a way that sensitive data and very customized processes are on premise, while the rest of the operation, which is not only standard but also consumes AI resources, is in the cloud. Moreover, finance, healthcare, and government are the industries that push more towards adopting hybrids as they can not transfer their whole data to the public cloud due to the regulations that they are subject to. From this, we infer that the future will not be defined by either of the options but will resort to the use of complex hybrid methods that would be able to carefully regulate access, adherence to standards, and, at the same time, benefit from the cloud, among others (NetSuite, 2024).

At the same time, the industry is moving towards platform convergence, where the traditionally different enterprise applications will eventually become one large digital ecosystem. The Business Technology Platform from SAP is a good example of this trajectory, as it is the shared services for integration, analytics, and AI deployment that the companies that are using the unified infrastructure for S/4HANA, SuccessFactors, and other applications are exploiting (SAP Investor Relations, 2024). However, on the other hand, the industry paradoxically shifts towards composable ERP architectures where the companies select the best-in-class components from different vendors and create their own solutions. Integration platforms like MuleSoft, Boomi, and Workato, as well as vendor-embedded capabilities such as SAP BTP, Oracle Integration Cloud, all enable organizations to interconnect the diverse systems and, at the same time, have consistency in the data (Panorama Consulting, n.d.). This trend of having two different paths simultaneously which is thus suggesting that the industry will be relying on hybrid approaches where the core ERP platforms not only integrate but also collaborate easily with the specialized applications, thus the flexibility of the ecosystem will be like that of the rigid suites.

The verticalization of the industry is speeding up as the vendors are now making specialized solutions along with the regulatory templates and functionalities that are tailored for particular sectors. For instance, SAP is offering different versions for more than 25 verticals, Oracle is creating industry clouds, and niche vendors are concentrating on single industries (Frandsen View Research, n.d.). The major vendors provide core platforms with industry overlays, while the minor ones are focused on deep specialization. Along with the consumerization of products, the entire redesign of user experiences is being done, where the main points are intuitive navigation, mobile-first design, and collaboration that is embedded. Workday's consumer-grade interface was the one that made user experience the competitive differentiator, and thus, it became unavoidable for SAP and Oracle to update their systems with the Fiori and

Redwood interfaces (NetSuite, n.d.). More and more, organizations are looking at the solutions in terms of user adoption instead of function completeness.

Pricing schemes have undergone noticeable changes, now encompassing not only traditional subscriptions but also consumption, outcome, and hybrid approaches. The trend indicates that value-based pricing is becoming more aligned with the company's achievements rather than being based on technical indicators (Panorama Consulting, n.d.). Low-code and no-code platforms are making customization available to everyone, and at the same time, the companies are putting money into the area of citizen development with solutions such as SAP Build, Oracle APEX, and Workday Extend. The industry is heading towards a scenario where two types of development take place: IT professionals work on the main platform while business users develop applications for their departments (NetSuite, n.d.; Panorama Consulting, n.d.).

Environmental, Social, and Governance skills are turning into core functions instead of extras. Suppliers integrate features such as carbon tracking, sustainable procurement monitoring, diversity analytics, and compliance management into their offerings (NetSuite, n.d.). Sustainability is thus positioned as an ERP dimension next to financial and operational management. The sector is moving towards systems that are fully automated and need very little human intervention. Oracle's self-driving database is a case in point, illustrating its features like self-tuning, self-patching, and self-repair (Oracle Corporation, 2025). This level of automation is applied not only to the execution of routine transactions but also to the implementation of process optimizations driven by performance data.

All these trends together point the advanced ERP landscape in the direction of a cloud flexible, AI-powered, and industry-specific platform that is delivered through business models with a customer-grade experience, autonomous operations, and integrated sustainability capabilities. The ERP sector is changing from being a transaction processing backbone to being an intelligent, adaptive, and sustainable digital core of the enterprise.

STRATEGIC ANALYSIS

The analysis of the strategy is performed by applying the theoretical frameworks to the empirical data obtained from the previous sections. The analysis focuses on the competitive dynamics, resource configurations, core competencies, and strategic choices of SAP, Oracle, and Workday in the advanced ERP industry. This part of the paper uses Porter's Five Forces model to gauge the industry's appeal, employs the VRIO framework to identify the strategic resources of the companies, recognizes the core competencies, studies the business level strategies, and finally looks at the corporate level and cooperative approaches.

Competitive Analysis for the Attractiveness of the Industry and the Five Forces Framework

Porter's Five Forces Framework offers a structured assessment of the factors that influence the attractiveness of the industry and the potential for profit. The model looks at the new entrants' threat, competitive rivalry intensity, the substitutes threat, the buyers' bargaining power, and the suppliers' bargaining power to evaluate the competitive pressures that define the strategic options in the advanced ERP industry.

Threats of New Entrants (Very Low) - The Advanced ERP market in its entirety is challenging, with the entrance of new players being quite hard and the existing vendors to maintain their market positions being quite easy. To create an AI-enhanced ERP platform that can compete, one would need to invest hundreds of millions of dollars at least, which would be applied to research and development, cloud infrastructure, sales and marketing, and implementation resources. Around 15% of the revenue is generated in R&D by SAP, hence one billion euros are invested into company's technology (SAP Investor Relations, 2024). Oracle, on the other hand, also spends a significant amount of money on research into autonomous computing and expansion of its data centers across the globe (Oracle Corporation, 2025).

The technological complexity which adds to the barriers of entry gets even tougher as the top line ERP systems use the power of AI and machine learning not only through but also in the company's financial management, supply chain optimization and human capital management, necessitating an expert team with a great deal of knowledge and skills in enterprise software architecture, machine learning engineering, natural language processing, and deep industry knowledge. Oracle's autonomous database technology, a product of years of engineering investment that competitors cannot easily match, is claimed to cost more than competitors, considering the cost over a long time period (Oracle Corporation, 2025). Incumbent advantages are more powerful due to network effects as machine learning models perform better with larger datasets, with SAP's algorithms gaining an edge from the data and insights generated across 440,000 customers (SAP Investor Relations, 2024). Customer switching costs are also a major factor in keeping incumbents in business since ERP implementations are very complex, take several years, and cost millions of dollars for large companies.

Competitive Rivalry (High) - The competition between established ERP vendors has become more contentious as companies try to gain a foothold in a slowly developing industry that is also very expensive to operate in and of great significance to the customers. The moderate concentration of the industry, where SAP has approximately 24% and Oracle 12%, generates conditions favorable for intensive competition (HG Insights, 2025). The AI features have turned into the main area of competition where companies spend a lot of money in order to make their own through machine learning, natural language processing, predictive analytics, and autonomous abilities.

By the year 2025, Oracle Corporation, through various statements, strongly positions autonomous technology as a critical differentiator and claims to obtain a significant operational cost benefit by implementing self-managing systems. On the other hand, SAP's stance involves full-fledged AI integration across the S/4HANA suite and the Joule copilot that functions as a conversational interface, thus making AI more and more accessible for the end user (SAP Investor Relations, 2024). Meanwhile, Workday focuses on 'naturally' integrated AI across its cloud-native applications and claims that this provides a better user experience (Reed, 2025). The shift to the cloud heightens the competition as the suppliers try to entice the customers who are already using their products and also to win over the new ones who are open to choosing the cloud-based solution. SAP's strategic concern here is to mitigate its vast user base from the old ECC system to S/4HANA while at the same time defending against the competitors who will seek to capture those users that are most vulnerable during the transition.

Threat of Substitutes (Low) - The ERP systems on the advanced level are extremely difficult to change due to full integration with all the business sectors, in-built AI functions, and their centrality to the operations of a large organization. The best pointy solutions need a lot of integration work to get the same data consistency and process coordination that ERP systems already provide as a native part of the integrated system. In most cases, it is too expensive for companies to develop their own systems because they would have to deal with such wide-ranging functionalities as financial management, procurement, supply chain, manufacturing, and industry-specific processes. Industry-specific platforms restrict themselves to only a part of ERP functions, which necessitates the connection of the financial and administrative systems, while the top vendors are more and more providing industry-specific versions that combine vertical functionality and wide-ranging ERP capabilities.

The high substitution barriers are created by the complexity and mission-critical nature of ERP, as the systems integrate with almost every other enterprise application, embed themselves into business processes, and possess years of transactional history. The introduction of AI capabilities, on the other hand, has the effect of reducing the substitutive risks since the value of on-board machine learning, predictive analytics, and autonomous operations is becoming very hard to imitate outside the all-encompassing ERP platforms.

Buyer Power (High) - Even after going through the expensive process of implementation, big enterprise customers still have the power to negotiate due to their enormous purchasing size and the nature of the market. Large firms commonly sign ERP contracts worth millions or even hundreds of millions of dollars, thus making it a procurement decision that gets noticed at the top management level and goes through a strict appraisal process. The companies put together a selection committee that includes members from various departments, with the committee sometimes lasting for years to evaluate different options, negotiate terms, and ask for proof of value. The big players like SAP, Oracle, Workday, Microsoft Dynamics, and even niche vendors are vying for the attention of large businesses with their solid solutions, thereby offering buyers a range of choices (HG Insights, 2025).

The level of buyer sophistication has been greatly improved as the companies gain experience with ERP implementations, hire specialized consultants, and get access to detailed product comparisons by means of analyst reports. Buyers want the AI capabilities to be shown in detail; they want to see the return on investment proofs through customer references, and they are more and more demanding of consumption-based pricing, which is in accordance with actual usage. Economic circumstances have a huge impact on the buyer's power since the uncertainty makes the buyers postpone the IT investments that are not crucial for their business and negotiate more aggressively on price and terms. Nevertheless, the buyer's power drops significantly after the implementation, as the switching costs go up to a great extent, and the organizations have already customized the systems, integrated the applications, and trained the employees.

Supplier Power (Moderate to High) - The advanced ERP market relies on multiple categories of suppliers who possess different levels of power. Among the suppliers, the cloud infrastructure providers, namely AWS, Microsoft Azure, and Google Cloud, are considered very important, with a lot of power using methods such as multi-cloud strategies or proprietary infrastructure like Oracle does (Oracle Corporation, 2025). AI computing resources, especially graphics processing units that are optimized for machine learning workloads, are among the suppliers with high power since NVIDIA is the major player in the AI computing market.

The need for specialized talent is probably the most prominent factor to be supplier power coming from the side of suppliers as manufacturers of advanced ERP development require people with expertise in various fields such as machine learning, natural language processing, autonomous systems, cloud architecture, and enterprise software engineering. The fight for such a skilled workforce is on the rise in the tech sector as nearly all the leading firms are rolling out AI features. The implementation partners, consisting of the major consulting companies, play a crucial role in providing important services that involve deploying ERP systems at the customer's premises. Here, both sides are somewhat dependent on each other as the vendors require partners that have the necessary capacity, while the partners rely on their relationships with the vendors. Figure 9 here shows a five forces diagram showing Threat of New Entrants (Very Low), Competitive Rivalry (High), Threat of Substitutes (Low), Buyer Power (High), Supplier Power (Moderate to High) and Figure 10 shows Porter's Five Forces analysis for the advanced ERP industry, showing the structural Dynamics shaping competitive intensity.

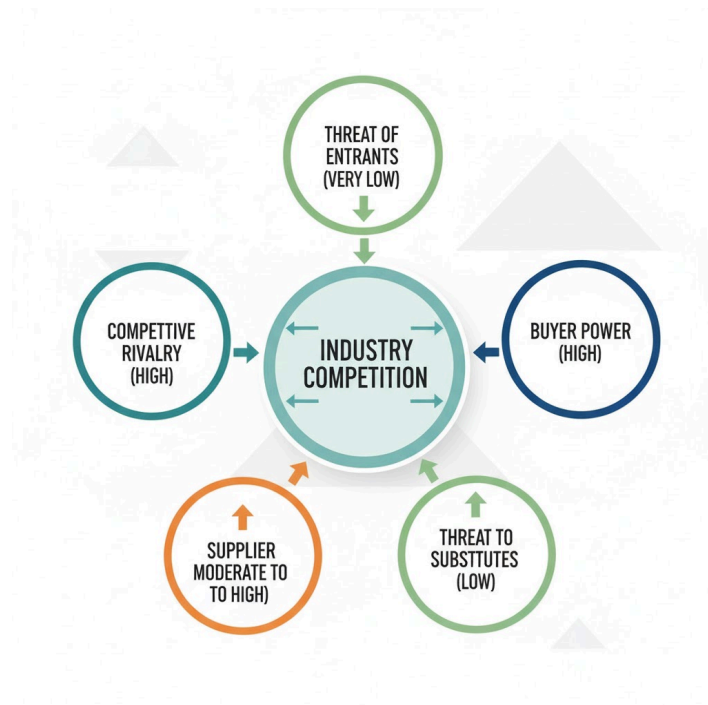


Figure 9, Five Forces Diagram showing Threat of New Entrants (Very Low), Competitive Rivalry (High), Threat of Substitutes (Low), Buyer Power (High), Supplier Power (Moderate to High)

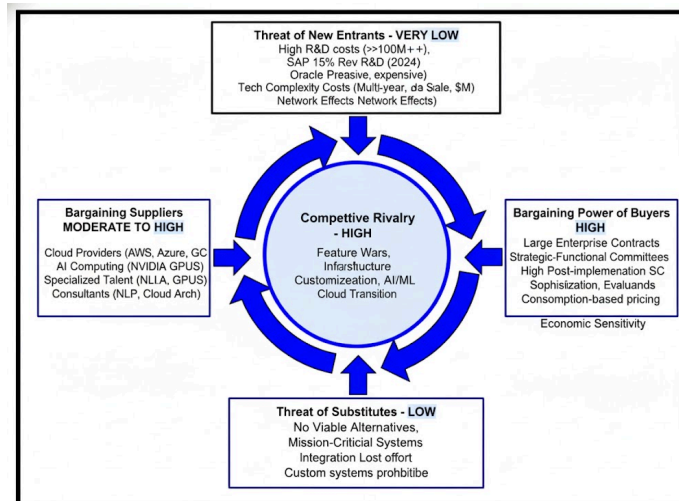


Figure 10, Porter's Five Forces analysis for the advanced ERP industry, showing the structural Dynamics shaping competitive intensity

The overall evaluation of the industry reveals moderate attractiveness among the factors mentioned above, notably high entry barriers safeguarding incumbents, fierce competition that pulls and pushes prices and forces the players to innovate, low threat from substitutes, the power of buyers that does not allow unreasonable prices to prevail, and moderate to high supplier power in critical resources.

Resource-Based View and VRIO Framework

The Resource-Based View argues that sustainable competitive advantage is based on valuable, rare, inimitable, and well-organized resources and capabilities. The VRIO model is a systematic approach to assessing how companies have resources that fit all four conditions to create sustained competitive advantage.

SAP VRIO Analysis - The S/4HANA Cloud platform, which contains AI and is based on in-memory computing, is the key resource that provides the competitive advantage that is long-lasting at SAP. The value of the platform lies in the fact that it allows real-time analytics that would have been impossible to implement with a traditional database architecture, predictive insights via the implementation of the machine learning system, and intelligent automation, minimizing human intervention. Customers have documented significant gains in terms of quick financial close, increased accuracy in demand forecast, and lower costs of transactions (SAP Investor Relations, 2024). Rarity is based on the proprietary HANA in-memory database of SAP that has performance features that competitors cannot achieve on conventional database platforms.

The causes of inimitability have several origins such as decades of organizational knowledge on business processes of enterprises in various industries and geographies, a rich patent base on in-memory computing and enterprise AI that establish legal barriers, 440,000-customer installed base that create network effects as machine learning models enhance with large population of data, and organizational culture, partner relationship, and brand reputation as socially complex assets that are hard to copy (SAP Investor Relations, 2024). SAP leverages these resources by investing in specific R&D teams, spending 15% of revenue on innovation, extensive partner and customer training and certification, business-specific solution

development units to satisfy vertical markets, and international support framework. This set of valuable and rare resources that are inimitable, combined with organizational capability, gives the company a long-lasting competitive advantage. Table 1 shows the VRIO Analysis for SAP and mentioning their their key resources.

	Resources	Valueable	Rare	Inimitable	Organization	Conclusion
	HANA database technology, embedded AI, and ML algorithms	S/4HANA Cloud with AI, in-memory computing, and real-time analytics	Proprietary HANA technology, AI integration throughout S/4HANA	50+ years of enterprise knowledge, extensive patents, 440,000 customer data advantage	15% revenue in R&D, global partner ecosystem, industry-specific units	Sustained Competitive Advantage
	Enterprise-focused ML models trained on business data	ML algorithms for predictive analytics, demand forecasting	Specialized algorithms for ERP processes	Accumulated training data, S/4HANA integration, process expertise	Embedded in modules, continuous improvement, and customer success programs	Sustained Competitive Advantage
	In-memory computing patents, enterprise AI applications	Patent portfolio in AI-enabled enterprise tech	Specialized patents for enterprise AI and ML	Legal protection, 50+ years of innovation accumulation	Dedicated IP management, strategic patent processes	Sustained Competitive Advantage
	€1B+ annual investment in innovation	R&D investment in AI/ML	High investment level (15% revenue)	Competitors can match spending levels	Effective R&D deployment, innovation culture	Temporary Competitive Advantage

Table 1, VRIO Analysis for SAP and their key resources

Oracle VRIO Analysis - Oracle's autonomous computing technology, which facilitates self-tuning, self-patching, and self-repair, is the most unique resource that gives it a sustained

competitive advantage. It is based on the fact that the operational costs of IT are significantly lowered due to systems being automatically managed without the manual control and enhanced reliability and performance. Companies that implement the use of the autonomous database developed by Oracle have recorded significant savings in administrative costs (Oracle Corporation, 2025). The reason is that Oracle is the sole large ERP vendor that provides truly autonomous systems in which infrastructure and application self-manage across the entire technology stack.

Inimitability is due to the decades of database technology leadership by Oracle that enables the autonomous capabilities, specialized algorithms that monitor and optimize systems in real-time, engineering culture that considers automation and autonomy, and architecture integration throughout the technology stack to make optimizations not possible by competitors based on multi-vendor architecture (Oracle Corporation, 2025). Oracle plans to take advantage of autonomous computers by positioning it as core differentiation, invest heavily in research and development to broaden its capabilities, customer success programs to show that it saves the company money, and vertical integration to ensure a smooth autonomous operation. This distinct capability is validated by the market through the 500 billion of ordered booked cloud (Reuters, 2025).

The vertical integration of Oracle between infrastructure and databases, up to AI-enabled applications, offers a long-term competitive advantage by creating synergies in the architecture that competitors cannot have. Performance optimization across the entire stack, simplified vendor management for the customers, combined security and compliance, and architectural benefits of AI workloads, such as the benefit of co-location of data, processing, and intelligence, are valued. The inimitability is due to the decades of database development that provided the basis, huge capital expenditure needed to construct the competitive infrastructure, company integration across infrastructure and application groups, and customer experience that has been established over the years across various product lines (Oracle Corporation, 2025). Table 2 represents the VRIO Analysis for Oracle and mentions their key resources.

	Resources	Valueable	Rare	Inimitable	Organization	Conclusion
	Autonomous database technology, AI automation across the stack	Autonomous computing (self-tuning, self-patching, self-repair)	Only major vendor with truly autonomous systems	Decades of database leadership, autonomous algorithms, and engineering culture	Aggressive marketing, substantial R&D, vertical integration	Sustained Competitive Advantage
	Oracle Cloud Infrastructure, Autonomous	Vertical integration infrastructure-to-application	Few vendors control the	Decades of database development, massive	Unified product roadmaps, integrated	Sustained Competitive Advantage

	Database, Fusion Cloud ERP	ications	complete technology stack	capital investment, and organizational integration	pricing, cross-layer optimization	
	Anomaly detection, automated journal entries, and intelligent cash positioning	AI models for financial automation	Specialized algorithms for financial processes	Training data from diverse customers, database integration, and PeopleSoft expertise	Embedded in Fusion Cloud ERP, continuous model improvement	Sustained Competitive Advantage
	40+ years of database engineering expertise	Database technology heritage	Moderate - other database vendors exist	Accumulated engineering knowledge, enterprise customer relationships	Dedicated database engineering teams, strategic positioning	Sustained Competitive Advantage

Table 2, VRIO Analysis for Oracle and their key resources

Workday VRIO Analysis - The cloud-based architecture of Workday, with machine learning integrated into it, allows the company to maintain a competitive advantage due to architectural benefits that other vendors, who have had to migrate legacy systems, do not have. The value is found in accelerated innovation processes because Workday has rolled out the features on an ongoing basis with no version fragmentation, better user experience through the modern design principles and architectural consistency since all customers are running on the same version, and easy integration of AI since it has been engineered as part of the platform since its beginning (Reed, 2025). Rarity is based on the fact that Workday is among the few large ERP vendors that are cloud-native by design, whereas SAP and Oracle have migrated their old architecture, inheriting the technical debt and design limitations.

The cause of inimitability is first-mover advantage since Workday designed its cloud-native architecture before other companies identified its significance. Organizational culture and engineering decisions to support continuous delivery are hard to copy by traditional vendors, customer expectations of persistent innovation generate switching costs, and decisions made at the time of initial platform design that can be modified with ease (Reed, 2025). Workday plans to take advantage of cloud-native architecture by using agile development models that allow fast delivery of features, a single product roadmap since all customers use the same

version, a customer success organization based on subscription retention, and strategic positioning that focuses on the merits of architecture. The 17 percent revenue growth per year is a sign of success in attracting customers who want an alternative that is modern (Reed, 2025).

The single HR-finance data model offered by Workday offers a sustained competitive advantage by being the only data model that offers unique data integration that is not available to vendors who have disintegrated systems. The value is based on the cross-functional AI insights, which find patterns in workforce and financial data, simplified reporting because data is in one structure, lower integration complexity, and more efficient machine learning because algorithms are trained using integrated data not siloed data. The inimitability is based on the design decisions of the architecture of an early platform that cannot be easily imitated by competitors and network effects, such that the Workday models get better when they learn through the unified customer base (ERP Cloud Training, 2025). Table 3 mentions VRIO Analysis for Workday and mentions their key resources.

	Resources	Valueable	Rare	Inimitable	Organization	Conclusion
	Multi-tenant design, single code base, continuous deployment	Cloud-native architecture with embedded ML	Few major vendors built cloud-native from inception	First-mover advantages, continuous delivery culture, and initial design decisions	Agile development, unified product roadmap, customer success for retention	Sustained Competitive Advantage
	Integrated HR and finance data structure	Unified HR-finance data model	Unique positioning spanning HR and finance	Initial architectural decisions, customer adoption patterns, and network effects	Integrated ML models, strategic positioning, cross-selling approach	Sustained Competitive Advantage
	Conversational interfaces, NLP for enterprise contexts	Natural language processing capabilities	Investment in enterprise-specific NLP	Accumulated training data, specialized models, and application integration	Integration across modules, continuous NLP improvement	Sustained Competitive Advantage

	Consumer-grade UI, mobile-first design	AI-driven user experience design	Modern interface differentiates	SAP Fiori and Oracle Redwood represent competitive responses	Design teams integrated with development, customer feedback processes	Temporary Competitive Advantage
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Table 3, VRIO Analysis for Workday and their key resources

Core Competence Analysis

Core competencies are the organizational strengths that bring about a competitive advantage due to their ability to combine organizational resources and coordination of various activities. In the case of advanced ERP vendors, core competencies should significantly contribute to customer perceived value, be hard to imitate by competitors, and lead to growth into the adjacent markets.

SAP Core Competencies - The main core competence of SAP is to combine AI and machine learning within the full scope of enterprise applications that are optimized with in-memory computing. This capability is reflected in S/4HANA being able to support huge volumes of transactions in real-time and at the same time execute sophisticated AI programs to make predictions, intelligent automation, and decision support. The competence also adds to the customer value as it helps organizations to work with real-time awareness, proactive insights, and automated processes to enhance their efficiency and quality of decisions (SAP Investor Relations, 2024).

This ability is so hard to achieve by competitors since it involves integrating various and constantly specialized skills, such as in-memory database engineering, machine learning algorithm development, enterprise process expertise in business functions, and organization coordination in thousands of developers and product managers. SAP has acquired this prowess after decades of experience in enterprise software, huge R&D investment, and experience of 440,000 customer implementations. The competence helps SAP to diversify into related areas such as intelligent planning and analytics, industry-specific clouds, sustainability management, and supply chain intelligence, proving it to be strategically significant to the company beyond core ERP (SAP Investor Relations, 2024).

Oracle Core Competencies - The core competence of Oracle is unique and focuses on autonomous computing, where systems automatically manage at the infrastructure, database and application levels. This competence incorporates database self-optimization algorithms that automatically tune the performance, automated patch management that applies the update without going offline, self-repair capability that identifies and fixes problems, and vertical integration through which it coordinates activities across the technology layers. The competence delivers customer value by generating dramatic cost reduction in operations, better reliability, better security, and optimization in performance (Oracle Corporation, 2025).

The competence is exceptionally hard to replicate by competitors, since it involves decades of experience in database engineering that Oracle has acquired, vertical control of infrastructure and applications that allows deep integration, a culture of commitment to automation throughout the engineering organization, and a large investment in the development

of autonomous algorithms. SAP and Workday are dependent on third-party providers of infrastructure, which does not allow them to build similar autonomous capabilities throughout the entire stack. Competence can also help Oracle diversify to autonomous data warehouse, autonomous transaction processing, and autonomous operating system, proving to be more than a core ERP application (Oracle Corporation, 2025).

Workday Core Competencies - The main core competence of Workday is the ability to seamlessly combine AI and machine learning to cloud-native applications and integrated HR-finance data models. This capability is seen through the fact that Workday has incorporated advanced machine learning algorithms in the entire platform without compromising user-friendly user experiences that enable non-technical users to access AI capabilities. The competence brings customer value in the form of predictive workforce analytics that forecast talent requirements, dynamically adjusted to actual results, intelligent financial planning, automated compliance monitoring, and integrated insights that connect workforce costs to financial results (Reed, 2025; ERP Cloud Training, 2025).

This competence is hard to emulate by competitors, as it needs cloud native architecture that is built to support AI-based deployment on a continuous basis, an integrated data model that blends HR and financial information, organizational culture that balances between AI complexity and user-friendliness, and experience in applying machine learning to talent and financial processes. Vendors with traditional vendors migrating legacy architectures also have a hard time reaching Workday consistency and deployment agility. The strength helps Workday to diversify into workforce planning and analytics, financial planning applications, procurement intelligence, and employee experience applications (Reed, 2025).

Generic Business-Level Strategy Analysis

The generic strategies framework by Porter establishes three methods used by firms in attaining competitive advantage, which include cost leadership, differentiation, and focus. The developed ERP market is characterised by competition based mostly on differentiation where the vendors differentiate based on their AI capabilities, technological excellence, and the breadth of their solutions instead of competing based on price as a priority.

SAP Business-Level Strategy - SAP also has a wide differentiation strategy that focuses on large organisations in different markets and offers end-to-end AI-powered ERP systems that are differentiated by technological advancedness, functional depth and ecosystems in implementation. Some of the sources of differentiation are in-memory computing technology that offers real-time processing power, embedded AI across S/4HANA offering predictive analytics and intelligent automation, industry-specific clouds which put vertical expertise into 25+ market segments, the numerous partner ecosystem which provides implementation capacity, and brand recognition built over decades of enterprise software leadership (SAP Investor Relations, 2024).

This wide differentiation approach is targeting the high-end market share, which includes customers who have a greater appreciation of a broad functionality, reliable service, global deployment availability, and a wide range of integration possibilities, rather than a low price. SAP prices high based on its advanced features, but the dominant policy demands massive persistence, research and development to ensure technological differentiation. This investment need is shown in the proportion of 15 percent of the revenue spent by the company in research and development (SAP Investor Relations, 2024).

Oracle Business-Level Strategy - Oracle focuses on a differentiation strategy that focuses on autonomous computing capabilities as the main distinguishing factor, with an aim of addressing the needs of enterprises which are interested in reducing their operational costs by adopting self-managing systems. Differentiation revolves around autonomous technology, which is autonomous (self-tuning, self-patching, and self-repair) through the entire technology stack, vertical integration which offers performance optimization, aggressive pricing (competitive total cost of ownership) and a huge investment in cloud infrastructure which allows it to be deployed globally (Oracle Corporation, 2025).

The strategy of Oracle is the combination of technological uniqueness with independent capabilities with competitive prices that build value propositions that attract customers with both innovative features and a favorable economic environment. This strategic positioning has been validated through the market by the 500 billion booked cloud orders (Reuters, 2025). The plan will focus on the customer base of SAP in the migration cycles when they are assessing alternatives and making autonomous capabilities and competitive price superior to the SAP value proposal.

Business-Level Strategy of Workday - Workday follows a targeted differentiation approach among certain customer groups such as Fortune 500 companies, educational organizations, health facilities and professional services firms with a modern cloud-native ERP characterized by excellent user experience, built-in AI, and an integrated HR-finance solution. The sources of differentiation are cloud-native architecture that allows unceasing innovation, new user interface with consumer-grade design principles, natural language processing that can be conversational interaction, unified HR-finance data model with cross-functional insights, and specific focus on HR and finance that allows the development of deep functional capabilities (Reed, 2025; ERP Cloud Training, 2025).

This specialized differentiation approach aims at those organizations that are more interested in user experience, speed of innovations, and cohesive talent-finance management, rather than in full breadth of functionality. Workday has chosen to limit scope to HR and finance instead of going the entire breadth of ERP, and thus invest heavily in user experience and functional depth. The 17% per annum increase in revenue is a manifestation of the efficacy of the strategy to attract customers who want modern solutions to the old systems (Reed, 2025).

Corporate Level and Cooperative Strategy Analysis

Corporate-level strategies address how firms manage their overall portfolio scope, resource allocation, and integration across business units. Cooperative strategies examine how firms leverage alliances, partnerships, and acquisitions to augment capabilities and expand market reach.

SAP Corporate and Cooperative Strategies - SAP has very limited vertical integration, resource allocation on application-layer AI with cloud provider infrastructure. Multi-cloud alliances with Azure, AWS, and Google Cloud can offer the global scale of infrastructure, the ability to have data centers regionally to meet sovereignty needs, as well as platform services to develop AI without the need to build capital-intensive data centers (SAP Investor Relations, 2024). This will enable SAP to focus its R&D resources in application-layer AI and machine learning algorithms and industry-specific solutions where differentiation takes place.

Diversification that is related adds variety to the enterprise applications of SAP with technological interconnectivity that is provided by the S/4HANA platform. Vertical knowledge is

brought into manufacturing, retail, healthcare, and financial services solutions by industry-specific clouds, which create switching costs and more enduring customer relationships. Human capital management is available on S/4HANA with SuccessFactors, which competes with Workday in a unified HR-finance. It is a diversification strategy that produces complete suites of enterprises that rival the monolithic vendors and the best-of-breed point solutions (SAP Investor Relations, 2024).

The rapid development of capabilities is achieved with strategic acquisitions, where Signavio is an AI-based process intelligence, Qualtrics is a customer and employee feedback analytics based on machine learning, and specific AI-based technologies are funded with startups. The strategies to collaborate are also aimed at an all-encompassing partner ecosystem where Microsoft Azure is the cloud platform of choice and SAP has a round of implementation partners who offer global deployment capabilities, along with an ecosystem of independent software vendors that develops an application ecosystem with thousands of complementary solutions (SAP Investor Relations, 2024).

Oracle Corporate and Cooperative Strategies - The most unique corporate strategy of Oracle is the wide scope of vertical integration, which manages all technology layers of infrastructure and down to database and applications. The foundation is offered by Oracle Cloud Infrastructure which includes compute, storage, networking, and security services that are optimized to run Oracle workloads. The Autonomous Database provides autonomous data platforms. The stack is fully filled with Fusion Cloud ERP and other applications (Oracle Corporation, 2025). This vertical integration facilitates autonomous computing differentiation of Oracle since the features of self-management are available at all technology levels, entailing immense amounts of capital expenditure on the construction of global data centers that cannot be attained by smaller competitors.

A great diversification in technology domains generates revenue streams in infrastructure services, database license, middleware, and applications as Oracle is able to provide consolidation of vendors where the customers use the entire stack Oracle provides. Key acquisitions established the Oracle ERP position, including PeopleSoft and JD Edwards who offered the customer base and intellectual property that served the base of Fusion Cloud ERP. Cerner makes Oracle get into healthcare featuring clinical systems and AI (Oracle Corporation, 2025). Selective alliances to enhance vertical integrations with Microsoft Azure interconnect to allow hybrid deployments and consulting firm relations as the source of capacity to implement.

Workday Corporate and Cooperative Strategies - Workday's limited vertical integration is a strategic choice that focuses all resources on improving application-layer innovation. The partnership with AWS provides global data center coverage, reliability, security, and platform services for AI development without requiring Workday to build and manage infrastructure (Reed, 2025). This strategy helps Workday maintain a focused organizational structure, concentrate research and development on HR and finance applications, and avoid heavy investments in infrastructure.

Restricting associated diversification allows Workday to remain concentrated on HR and finance, as well as enter other related fields like planning and analytics, procurement, and employee engagement. This lack of diversification enables the functional capability to be developed deep in special areas rather than the resources being spread too thin and across a broad spectrum of ERP functions. Strategic acquisitions speed up capability growth, with Adaptive Insights adding AI-powered planning, Peakon introducing machine learning for employee

engagement, and HiredScore offering AI for talent acquisition (Reed, 2025). The most significant working relationship of Workday is the AWS infrastructure partnership, which is supported by implementation partnerships with major consulting companies, technology partnerships on specialized AI tools, and university work on machine learning. Table 4 shows Comparative Corporate and Cooperation Strategies of Advanced ERP Leaders.

Strategic Dimension	SAP	Oracle	Workday
Vertical Integration	Limited - application layer and HANA database, partners with Azure, AWS, Google Cloud for infrastructure	Extensive - infrastructure through application ; proprietary Oracle Cloud Infrastructure	Minimal - application focus only AWS partnership for infrastructure
Diversification	Related diversification - 25+ industry clouds, SuccessFactors HCM, Analytics Cloud, sustainability management and procurement	Broad across technology domains: infrastructure, databases, middleware, ERP/HCM/CX applications, industry solutions	Limited related: core HR and finance, planning and analytics, procurement, employee engagement
Major Acquisitions	Signavio (2021) - process mining and AI intelligence, Qualtrics stake (2018) - experience management with ML, ongoing AI startup investments	PeopleSoft (2005) \$10.3B for ERP and HCM, Cerner (2022) - healthcare AI, DataFox (2018) - AI data enrichment	Adaptive Insights (2018) - AI planning; Peakon (2021) - ML employee analytics, HiredScore (2021) - AI talent acquisition, Scout RFP (2019), VNDLY (2020)
Strategic Alliances	Microsoft Azure preferred cloud partner, AWS and Google Cloud partnerships, Accenture, Deloitte, PwC, EY, KPMG implementation, thousands of ISV partners	Microsoft Azure interconnect for hybrid deployments, Accenture, Deloitte, PwC, EY, KPMG implementation; thousands of ISV partners	AWS as primary infrastructure provider; Accenture, Deloitte, PwC, EY implementation, technology partnerships for NLP and AI, academic collaborations for ML research

Table 4, Comparative Corporate and Cooperation Strategies of Advanced ERP Leaders

RESULTS AND CONCLUSION

The strategic analysis of the advanced ERP industry gives significant information on the competition between SAP, Oracle, and Workday in the artificial intelligence-based enterprise software market. The use of Porter's Five Forces, VRIO analysis, core competence evaluation, and strategy frameworks all contribute to understanding the competition and the factors that support the industry's sustainable advantage.

The advanced ERP sector is moderately attractive, with its features safeguarding and, at the same time, rewarding the creativity of the new companies. The newcomers are subjected to extremely high barriers since the creation of a competitive AI permitted ERP platform alone costs hundreds of millions of dollars, requires good technological knowledge, and the latter years of experience in the field. A huge customer base of 440,000 for SAP and database expertise for Oracle creates a situation that new players cannot quickly copy, which in turn explains why 24%, 12%, and 8% market shares, respectively, are held by SAP, Oracle, and Workday. The principled conflict among these aged companies continues to be very intense, as the dominance of AI techs has turned out to be the major point of competition. SAP employs AI throughout the whole life of S/4HANA with the help of Joule assistant, Oracle puts forward the technology that manages itself as the main distinguishing factor, and Workday's AI is claimed to be a seamless integration with human users to ensure user friendliness. The large enterprise customers have considerable power as the multimillion-dollar contracts allow them to conduct a thorough evaluation of several vendors. The supplier's power ranges from moderate to high, where the most limiting factor is the specialized talent because there is fierce competition for machine learning experts. The threat of substitutes is still low as the comprehensive integration and embedded AI of the advanced ERP system are so superior that it's hard to replicate.

According to the VRIO analysis, all the companies get to possess valuable, rare, difficult to imitate, and well-organized resources that give them sustainable advantages, even though the sources are quite different among them. SAP's S/4HANA Cloud, with its built-in AI, HANA in-memory computing for real-time analytics, and machine learning algorithms fed with customer datasets of 440,000, and the aggressive patenting tactics give it the upper hand. The 50 years of expertise, the installed base that produces data advantages, and the company's culture make this process hard to imitate. SAP benefits from these resources through its 15% revenue research investment, a wide range of partner ecosystems, and a global support infrastructure. Nonetheless, the company is likely to encounter difficulties during the ECC to S/4HANA migration as rivals might focus on the customers, the cloud native competitors might attract with better user experiences, and Oracle's pricing policy might threaten the company's leadership in the market.

The unique strength of Oracle is that autonomous computing is the technology that enables self-tuning, self-patching, and self-repair of both infrastructure, database, and applications, which significantly lowers the cost of IT and enhances reliability. Oracle is the first and only major vendor that provides fully autonomous systems across the technology stack. This is hard to replicate by the decades of expertise in databases, specialized algorithms, engineering culture of automation, and the integration architecture that competitors cannot match. Integration of infrastructure up to application offers further benefit via optimization of performance and

easier dealing with vendors. The booked cloud orders of 500 billion are an indication of market validation. Nonetheless, Oracle has fewer installed bases that help to create network effects, vertical integration that demands immense amounts of capital, and must keep on proving independent value.

Cloud-native architecture that includes in-built machine learning offers Workday sustained benefit, increasing their innovation cycles, a better user experience, and a smooth AI integration that is developed from the very beginning. The single HR-finance data model is beneficial because the cross-functional AI insights are not available to separate systems. NLP allows conversational interaction, distinguishing the user experience. These are hard to replicate since Workday developed the cloud-native architecture before its competitors understood the significance of cloud-native architecture and the organizational culture that maximizes on the constant delivery. The annual revenue growth of 17 percent proves that the markets have been validated. Nevertheless, Workday has a narrow scope that limits its reach to the market, bigger competitors who invest in the improvement of the user experience, and reliance on a particular customer group.

These three companies demonstrate that machine learning should be applied in all applications, and not as a supplement, since it is the fundamental competency that should be created, mastery of an enterprise, and organizational alignment. This is because success involves bringing about massive changes, and SAP has exhibited this by having thousands of implementations that have taught them to do the same. The sector requires innovation that is always ongoing, with SAP spending 15 percent revenue on research, Oracle coming up with autonomous technology, and Workday providing speedy delivery. To be successful, it is necessary to balance between standardization and customization with solutions such as SAP BTP, the industry cloud of Oracle and Workday extensions.

Differentiation-based competition is demonstrated at the business-level strategy. SAP is broadly differentiated to target large enterprises with wide-ranging AI-based solutions in the in-memory computing, embedded AI, industry clouds in 25+ verticals, vast partners, and brand recognition. This demands high prices that could only be achieved by carrying out extensive research. Oracle is differentiating itself based on independent computing and competitive pricing, which has been substantiated by 500 billion cloud orders. Workday is working towards focused differentiation in the Fortune 500, education, healthcare, and professional services sectors, utilizing a cloud-native ERP that delivers a better user experience, built-in AI, and integrated HR-finance functionality, resulting in 17% revenue growth.

Strategies at the corporate level are different. Oracle aims to have the vast vertical integration that governs infrastructure with applications so that it can have autonomous capabilities but it demands enormous capital investment. SAP embodies a limited scope of vertical integration, emphasizing applications, and collaborates with Azure, AWS, and Google Cloud so that it can concentrate on application AI research. Workday is following a low vertical integration model that attempts to be application-oriented and collaborate with AWS, which allows organizational emphasis but results in infrastructure addiction. The strategies of diversification vary as SAP approaches related diversification with enterprise applications, Oracle pursues broad diversification with technology, and Workday pursues limited diversification with a focus on HR and finance. SAP has the best sustained competitive advantage, standing on various factors that reinforce the competitive advantage. The company has its precious and unique resources, such as S/4HANA along with inbuilt AI, algorithms that

are trained on almost 440,000 customer records, vast patents, and 50 years of experience. The key skills in the process of AI integration and the engines of change allow achieving success in the largest business segment. Broad differentiation that demands premium pricing is compatible with capabilities and diversification based on platform investments and partnerships, which can provide implementation capacity. The validation is seen in the 24% market-share and in 87 percent of Forbes Global 2000 list of corporations. Nonetheless, SAP needs to carry out cloud migration and protect against attacks, support HANA benefits, develop artificial intelligence features, and upgrade user experience.

Oracle has good positioning in terms of autonomous and vertical integration. High imitation barriers are generated by resources either in the form of database expertise, investments in infrastructure and integration of architecture. Competencies facilitate differentiation of operating costs, whereas strategy of combining autonomous capabilities and competitive pricing form its strong propositions. The cloud orders of \$500 Billion are indicators of successful implementation. Nevertheless, the Oracle has fallen behind in terms of installed base pursuing network effects, vertical integration that takes colossal investment, competitive partner tensions as well as the necessity of proving autonomous value.

Workday exhibits specific positioning in the form of cloud-native structure, integrated data model, and excellent user experience. The resources are hard to copy because the conventional vendors cannot cope with the cloud-native benefits. Competencies facilitate leadership between segments that place more emphasis on experience than integrated functionality. The 17% growth in revenue area is effective. Nonetheless, Workday has a narrow scope to restrict the market, to competitors with a better user experience, segment dependence, and the necessity to innovate continuously.

According to the application of the framework, it is most probable that SAP will be able to sustain competitive advantage because the scale of the installed base will offer network effects, full functionality to meet complex demands, the presence of a large ecosystem, decades-long brand name, and significant investment in research. Though Oracle autonomous capabilities and Workday cloud-based architecture are highly differentiating, the accrued strengths of SAP across several planes are the hardest to counterfeit. But to be sustainable, it requires doing cloud transition, leading in technology, and protection against competitive attacks.

A number of imperatives come out in the sustainable advantage. To start with, advantage demands resources, which are not only valuable but are also rare, hard to imitate, and also exploited using proper structures. Second, to succeed, competencies that combine AI, arrange changes, and innovate constantly and standardization and customization should be balanced. Third, corporate strategies should be in line with resources and capabilities. Fourth, sustained benefit is a result of cumulative benefits such as installed base, organizational knowledge, ecosystem relationship and brand reputation. Fifth, the development of the industry needs the active course in many directions at once, to seek both universal platforms and solutions and specialized systems and control over vendors as well as openness of the ecosystem.

The future of the industry will be characterized by further still competition and the market leadership will be based on the ability of the firms to survive evolution to cloud-flexible and AI-powered solutions with at the same time preserving unique benefits. The challenge at SAP is to migrate installed bases at the expense of maintaining leadership. The issue facing Oracle is to show independent value and to keep growing. The challenge facing Workday is to increase the addressable market at the same time with benefits. To be successful, it is crucial to

be a consistent innovator with huge investment, efficient implementation, and a strategic decision that is in line with unique resources, capabilities, and position. Companies that best use strategic frameworks to interpret the nature of competitive forces, find the bases of advantages, build competencies, and make consistent decisions will have a disproportionate value as the industry becomes the intelligent, adaptable, and sustainable digital backbone of most businesses.

REFERENCES

- Apps Run the World. (2024). *Top 10 ERP Software Vendors and Market Forecast 2024–2029*. <https://www.appsruntheworld.com/top-10-erp-software-vendors-and-market-forecast>
- Barney, J. (2011b). Resource-based theory and vertical integration. *Strategic Management Journal*, 32(4), 397–420.
- Barney, J. (2011c). Strategic alliances and cooperative advantage. *Academy of Management Perspectives*, 25(3), 56–69.
- Bradford, M. (2015, January 1). *Modern ERP: Select, implement, and use today's advanced business systems* (2nd ed.). Lulu Press. <https://www.lulu.com/>
- Burson, J. (2025, October 2). *SAP vs. Oracle ERP comparison*. Paperfree. <https://paperfree.com/en/magazine/sap-vs-oracle-erp-comparison>
- Cloud ERP market size, share and Growth Report, 2030*. Cloud ERP Market Size, Share And GrowthReport, 2030. (n.d.). <https://www.grandviewresearch.com/industry-analysis/cloud-enterprise-resource-planning-erp-market-report>
- Ciobanu, A. (2025). *A Longitudinal Study of ERP System Capabilities and Value*. <https://doi.org/10.2139/ssrn.5200609>
- Davenport, T. H. (1998). Putting the enterprise into the enterprise system. *Harvard Business Review*, 76(4), 121–131.
- ERP Cloud Training. (2025, August 27). *Workday Financials vs. SAP: An in-depth analysis*. <https://www.erpcloudtraining.com/blog/workday-financials-vs-sap>
- Fortune Business Insights. (2024). *ERP Market Size & Leaders in 2024*. The CFO Club. <https://thecfoclub.com/operational-finance/erp-market-size/>
- Garg, P., Gupta, B., & Dwivedi, Y. K. (2023). Examining the role of artificial intelligence in enterprise resource planning (ERP) systems: A future research agenda. *Journal of Enterprise Information Management*, 36(6), 1–28. <https://doi.org/10.1007/s43088-023-00460-y>
- Gartner. (2023). *Magic Quadrant for Cloud ERP for Product-Centric Enterprises*. Gartner Research.
- Global Market Insights. (2024). *ERP software market report*. GMI.
- Goovaerts, D. (2025, July 1). *Oracle poised for explosive growth with \$30B cloud deal*. Fierce Network. <https://www.fierce-network.com/cloud/oracle-poised-explosive-growth-30b-cloud-deal>

- Haug, A., Arlbjørn, J. S., & Pedersen, A. (2010). The role of ERP systems in inter-firm cooperation. *Industrial Management & Data Systems*, 110(5), 747–763. <https://doi.org/10.1108/02635571011044767>
- Haug, A., Pedersen, A., & Arlbjørn, J. S. (2010). ERP system strategies in parent-subsidary supply chains. *International Journal of Physical Distribution & Logistics Management*, 40(4), 298–314. <https://doi.org/10.1108/09600031011045316>
- Hayes, M., & Downie, A. (2025, April 17). *Artificial Intelligence in ERP*. IBM. <https://www.ibm.com/think/topics/ai-in-erp>
- HG Insights. (2025). *ERP market share, size & key players in 2025*. <https://hginsights.com/blog/erp-market-share-size-report>
- Jacobs, F. R., & Weston, F. C. (2007). Enterprise resource planning (ERP) — A brief history. *Journal of Operations Management*, 25(2), 357–363. <https://doi.org/10.1016/j.jom.2006.11.005>
- Jawad, M., & Balaczs, A. (2024). Innovation and differentiation in ERP systems: Strategic shifts in enterprise technology. *Journal of Information Systems and Technology Management*, 21(2), 155–170.
- Matoso, J. O., & Abib, G. (2015). Competitive advantage of a franchisee in the software market: The case of totvs. *Journal of Information Systems and Technology Management*, 12(3), 627–646. <https://doi.org/10.4301/s1807-17752015000300007>
- Matoso, L., & Abib, G. (2015). Strategic alliances and acquisition strategies in the software industry. *Management Research Review*, 38(6), 626–642. <https://doi.org/10.1108/MRR-09-2014-0219>
- Mhaskey, S. V. (2024). *Integration of Artificial Intelligence (AI) in Enterprise Resource Planning (ERP) systems: Opportunities, challenges, and implications*. *International Journal of Computer Engineering in Research Trends*, 11(12), 1–9. <https://doi.org/10.22362/ijcert/2024/v11/i12/v11i1201>
- NetSuite. (2024, September 27). *60 critical ERP statistics: Market trends, data and analysis*. <https://www.netsuite.com/portal/resource/articles/erp/erp-statistics.shtml>
- Pokala, S. (2024). The evolution of ERP: A systematic review of AI and cloud integration. *Journal of Technology Management & Innovation*, 19(2), 45-58.
- Porter, M. E. (1980). *Competitive strategy: Techniques for analyzing industries and competitors*. Free Press.
- Porter, M. E. (1985). *Competitive advantage: Creating and sustaining superior performance*. Free Press.
- Rashid, M. A., Hossain, L., & Patrick, J. D. (2002). The evolution of ERP systems: A historical perspective. In L. Hossain, J. D. Patrick, & M. A. Rashid (Eds.), *Enterprise Resource Planning: Global Opportunities and Challenges* (pp. 1-16). Idea Group Publishing.
- Reed, S. (2025, September 17). *Workday's growth trajectory and competitive positioning: Long-term value creation in the cloud ERP sector*. AInvest. <https://www.ainvest.com/news/workday-growth-trajectory-competitive-positioning-long-term-creation-cloud-erp-sector-2509/>

- Reuters. (2025, September 9). *Oracle expects half a trillion dollars in booked cloud orders, stock rises 27%*. Reuters. <https://www.reuters.com/business/oracle-expects-half-trillion-dollars-booked-cloud-orders-stock-rises-27-2025-09-09/>
- Tech Research Firm. (2024). *ERP Software Market Growth Analysis*. Tech Research Firm.
- Grand View Research. (n.d.). *ERP software market size, share & trends analysis report*. Grand View Research. <https://www.grandviewresearch.com/>
- NetSuite. (n.d.). *Enterprise Resource Planning (ERP): Overview and industry trends*. NetSuite. <https://www.netsuite.com/>
- Oracle Corporation. (2025, September 9). *Oracle Announces Fiscal Year 2026 First Quarter Financial Results*. <https://investor.oracle.com/investor-news/news-details/2025/Oracle-Announces-Fiscal-Year-2026-First-Quarter-Financial-Results/default.aspx>
- Pang, A., & Markovski, M. (2024, December 24). *Oracle on track to lead both cloud ERP and total ERP. Apps Run The World*. <https://www.appsruntheworld.com/oracle-on-track-to-lead-both-cloud-erp-and-total-erp/>
- Panorama Consulting. (n.d.). *2024 ERP report: Trends, challenges, and best practices*. Panorama Consulting Group. <https://www.panorama-consulting.com/>
- Pinjala, S., Roy, R., & Seetharaman, P. (2016). Firm growth and innovation in the ERP industry: A systems thinking approach. *arXiv*. <https://arxiv.org/abs/1606.03539>
- SAP Investor Relations. (2024). *SAP annual report 2024*. SAP SE. <https://www.sap.com/investors.html>
- Singh, A. V. (2025, April 28). *SAP surpasses Salesforce, Oracle, and Workday in cloud revenue growth*. TechGig. <https://content.techgig.com/technology/sap-surpasses-salesforce-oracle-and-workday-in-cloud-revenue-growth/articleshow/120697761.cms>
- St. John, C. (2024, October 17). *Top 14 cloud platforms — overall comparison, market share*. Medium. <https://medium.com/@csjcode/top-14-cloud-platforms-overall-comparison-market-share-d788c4b9e877>
- TechGig. (2024). *SAP surpasses Oracle, Workday, and Salesforce in cloud ERP growth*. TechGig. <https://content.techgig.com/>
- Zwets, B. (2025, April 18). *Analysis: Oracle beats SAP in ERP market*. Techzine Global. <https://www.techzine.eu/news/applications/130690/analysis-oracle-beats-sap-in-erp-market/>